

aTOMos v3.6.1.10

Orbital seeds and timestamp queries

Ground-station horizons with the literal self-referential kernel

MEO / GEO / inclined geosynchronous / high elliptical
Two ASA/NA sets · synchronous JK · CPU / CUDA

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Preserved parent	3.6.1.9 ORBIT-SEED-R1
Target	RTX 5070 Ti Laptop / SM120 / FP64
Release	3.6.1.10 / 15 September 2026

8.51–8.86 KiB per complete demonstration seed

Reconstruct physical satellite states at requested timestamps. Keep full state, time, Up and model dependencies. Measure forecast error against later reference orbit data and report when each accuracy budget is exceeded.

Editable equations, native binaries, packed examples, a local timeline interface and verified evidence.
Prepared with AI assistance.

Release and implementation record

This subversion implements **ORBIT-SEED-R1**. It packs a complete physical model and reconstructs a satellite's ground-station-relative position at an arbitrary supported timestamp. The examples cover real MEO, GEO, IGSO and high elliptical targets. Two explicit ASA/NA mask sets feed the existing synchronous JK operation; the resulting state changes the ordered query schedule.

Delivered capability	Concrete implementation
Compact seeds	8,716–9,074 bytes, including all per-seed physical, frame, forcing, station, query and word parameters.
Timestamp scrubbing	Native CPU canonical checkpoints, bounded memory, exact fractional query time and query-order-independent states.
Physical prediction	Full degree/order-12 Earth gravity, Sun/Moon forcing, radiation pressure, fitted RTN acceleration, varying EOP and Schwarzschild correction.
Literal state	Two ordered whole-word ASA/NA stages, editable X/J/K equations, synchronous JK and a trace of the selected next timestamp.
Station geometry	Full ECEF/ENU position and velocity, Up, range/range rate, azimuth/elevation, original UGTS keys and complete time/winding.
Both horizons	Refined visibility events and sampled prediction-error horizons against withheld later orbit data.
Execution evidence	Native CPU/CUDA, independent equations and propagation, isolated packed-only replay, physical reference comparisons and actual-device sanitizer.
User interface	Local timeline, exact timestamp input, satellite/station selection, pass search and ordered feedback replay.

The seed epoch is 2 January 2025. A fixed rule chooses among one-, three- and seven-day earlier training windows using an earlier validation day. R2 keeps all four January examples below 10 metres at every tested first-day reference sample. Longer intervals still have failures. January is a reused development benchmark; a separate February confirmation follows in the evidence. The default codec uses lossless one-bit planes in machine words. Propagation still uses binary64 arithmetic, with measured CPU/GPU and independent integration differences.

The prior LIVE-GPS-L1-R1, CGK-R1, SRK-R1 and SATNAV-R1 equations remain in the second part of this volume. Their previous receiver-position measurements are historical component evidence, not satellite forecast accuracy. The inherited 3.6.1.9 measurements are distinguished from the new 3.6.1.10 checks in the optimization chapter. The parent release and its delivered files are preserved separately.

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1 Seed execution optimization in 3.6.1.10

This release applies the usable word-organization ideas from the newly supplied 21-page design document to the complete orbital seed. The source audit in the next section distinguishes the source's equations from the corrected executable adaptations. The preserved parent is 3.6.1.9. The seed schema remains ORBIT-SEED-R1, schema version 3.6.1.9; release and schema versions are deliberately separate so existing seed bytes and content identities remain valid.

1.1 Exact six-stage word transpose

Let w_r be one of 64 numeric bit patterns and let $a_{r,c} = ((w_r \gg c) \& 1)$ for $0 \leq r, c < 64$. The storage operation is the bit-matrix transpose

$$p_c = \sum_{r=0}^{63} a_{r,c} 2^r.$$

Both the old and optimized implementations produce exactly these words. The new implementation places the block in an exact multiword temporary, $V = \sum_r w_r 2^{64r}$. For each $s = 32, 16, 8, 4, 2, 1$, define

$$M_s = \sum_{\substack{r \& s = 0 \\ c \& s \neq 0}} 2^{64r+c}, \quad d_s = 63s,$$

and perform the simultaneous exchange

$$T = (V \oplus (V \gg d_s)) \& M_s, \quad V \leftarrow V \oplus T \oplus (T \ll d_s).$$

Each stage exchanges one row-index bit with its corresponding column-index bit. The six exchanges transpose the complete block. Applying the transpose twice is the identity. This is a binary permutation, not a Hadamard transform of the satellite coordinates. Python's integer is an exact multiword temporary; the serialized representation still uses little-endian uint64 words.

The codec retains plane-major order, numeric type tags, zero tail padding, canonical byte encoding and SHA-256 content identity. Negative zero, subnormal binary64 values and signed/unsigned integer limits retain every bit. Both legacy canonical and bit-plane codecs remain readable. The public encoder and decoder now share the 300,000-word count limit.

1.2 Measured packing improvement

An independent mapping check covers all 4,096 basis bits. Because every operation is linear over GF(2), the basis check establishes the specified mapping for every block. Random tests also cover 137 lengths, including partial blocks and transitions between groups. All eight January/February example seeds repack to exactly their original bytes and retain their physical-model hashes.

Operation	Parent median	New median	Ratio
Words to planes	5.858 ms	0.228 ms	25.72
Planes to words	6.167 ms	0.145 ms	42.55
Complete seed pack	8.783 ms	3.162 ms	2.78
Complete seed unpack	17.103 ms	5.528 ms	3.09

These are local host measurements for the C03 seed, with nine measurements of five calls after warmup. Full ranges, Python version, source hashes and seed identities are recorded in `results/seed_word_optimization/benchmark_final.json`. The ratios describe packing and loading; they are not claims about orbital integration throughput. Seed sizes and physical prediction errors do not change as a result of this permutation.

1.3 Prepared queries with consistent state identity

The GCRS-to-ECEF query evaluates the frame and derivative together, $\mathcal{F}(t) = (M(t), \dot{M}(t))$, sharing the same precession, Earth rotation, polar motion and polynomial evaluations. The resulting operations retain the parent's arithmetic ordering. Station ECEF origins and ENU matrices use a bounded cache keyed by the exact binary64 latitude, longitude and height bits. Parsed Boolean inputs use a bounded cache keyed by canonical feedback content. Signed zero and edited values cannot alias a different cached input.

A worker holds the physical model snapshot used to create its numeric transport. A session holds one complete seed snapshot and checks its physical-model identity against the worker. Public seed/model views are read-only. To edit, use `session.copy_seed()` or `worker.copy_model()` and construct a new session or worker. A station or feedback edit can reuse the same worker when the physical model is unchanged. This repairs the prior possibility of using an edited Python frame with an older native model or a stale trace identity.

With one fixed parent binary/adaptor to isolate Python preparation, five-repeat median measurements give 1.15–1.29 times faster 256-query workloads and 1.20–1.38 times faster first-day schedules across the four objects. Complete mathematical outputs match the parent exactly. This comparison does not claim a whole-release cold integration speedup. See `results/query_overhead_comparison_final.json` and `results/query_snapshot_independent_check.json`.

1.4 Native domain, surface and repeated-query behavior

ORBIT-SEED-NATIVE-3 carries the dynamics kind and exact declared time interval before the existing numeric model fields. Versions 1 and 2 remain readable; their domain is the intersection of the required forcing/frame series. The new worker rejects queries outside the declared model interval, inconsistent epochs and physically undefined propagation at or below the reference Earth surface. Python already rejected Earth-penetrating acceleration; the old native path could return it as a valid state. These checks now agree.

For the canonical integration lattice $t_n = nh$, each RK4 call receives its exact indexed endpoint t_{n+1} , or the exact requested fractional endpoint. This avoids rejecting a valid closed boundary when floating-point evaluation of $nh + h$ differs from $(n + 1)h$. Midpoint labels stay inside that endpoint interval. The existing 30 s and 7.5 s frozen model grids retain their numerical states.

Within a batch, timestamps are keyed by their exact binary64 representation, including the distinction between positive and negative zero. First occurrences are propagated; later identical entries reuse the result and preserve input order. For 64 copies of the 300 s test timestamp, actual RK4 work decreases from 640 steps to 10. Reused rows report zero additional steps and an explicit reuse flag. CUDA still pays launch/upload overhead; this work reduction is not a claim of a 64-fold wall-clock improvement.

1.5 Fresh execution and compatibility evidence

The CPU replay reconstructs all 15,553 archived January/February forecast positions exactly, at the original reference timestamps. All eight first-day reference comparisons remain below the adopted 10 m engineering budget. Packing itself adds zero numerical error. The eight complete seed files remain 8,716–9,147 bytes across both dates and retain their original hashes.

The native audit compares 1,040 actual CUDA trajectories and 256 complete word transitions, plus 108 cache replay queries. The largest sampled CPU/CUDA position difference remains $4.31961135 \times 10^{-5}$ m, and velocity difference remains 8.07049×10^{-9} m/s. Compute Sanitizer reports zero errors for its 129-query trajectory/word workload. Both CPU and CUDA builds pass all four CTests. The broader archived CUDA replay checks 15,553 epochs: all 8,064 January

predictions match the parent CUDA positions exactly. February's 7,489 comparisons use the CPU baseline and reach 0.220015 mm maximum difference, within the 10 mm CPU/CUDA contract. This broader sample set is distinct from the 1,040-query audit above. The fresh packed-only replay denies reads of the original release, exercises both ASA/NA sets and a counterfactual JK equation, and refines seven-day events for all four satellites at both stations. Independent retained SATNAV verification checks 256 epochs and 4,828 fields.

The final machine-readable index `validation_results.json` binds the new runtime files, binaries and individual reports. It distinguishes inherited physical-reference evidence from these new execution checks. Detailed old investigation folders retain their original status and version; they do not supersede the final index.

1.6 Validation scope

The following physical equations and January/February accuracy measurements were established in 3.6.1.9. They remain baseline evidence for the same frozen models. The new release checks serialization, native execution, replay and query results separately. An exact pack does not remove floating-point integration error or unknown future manoeuvres. The previously reported longer-horizon failures and missing reference intervals remain part of the measured system limits.

1.7 Supplied phi source: exact transfers and corrections

The useful transfer is an explicit finite-state, bitfield architecture. R10 applies an exact bit permutation to its existing lossless bitplane format. The supplied text does not establish replacement orbital dynamics or a more accurate geometric bearing. Its claimed properties must follow from its equations, separately from the names given to them.

Provenance. The supplied 21-page PDF is an exported Google Search AI conversation, with pseudocode and mathematical claims. References here use its visible physical page numbers, checked against rendered pages. The source is `C:/TOMONETMALFORMED/3.6-phi-2pi-fibonacci-36-10---07---1990-gregorian-calander-Tom-Klootwijk.pdf`. Its SHA-256 is `0fddb55ba01bad0c94b435ddf89c4b778e8a94776946f6cc60cfbed860c5116`. The full decision record is `docs/PHI_SOURCE_AUDIT.md`. No future target observations or changed force parameters enter this transfer.

Metric and information counterexamples. For source page 5, let $\phi = (1 + \sqrt{5})/2$, $H_4 = H_2 \otimes H_2$ and $A = \text{diag}(\phi, \phi^{-1}, 1, -1)H_4$. Then

$$H_4^T H_4 = I, \quad |\det A| = 1, \quad \|A(1, 0, 0, 0)^T\|_2^2 = \frac{\phi^2 + \phi^{-2} + 2}{4} = \frac{5}{4}.$$

Thus A preserves four-dimensional volume magnitude, not Euclidean lengths or all areas. Page 9's integer implementation also maps both $(0, 0, 0, 0)$ and $(-1024, -1024, 0, 0)$ to zero after its masks: the second input has sums $(-2048, 0, -2048, 0)$. Both inputs fit the declared fields. This arithmetic cannot serve as a lossless physical-state encoding.

Executable corrections. Page 8's conditional mask is always zero because a right shift by 63 yields only 0 or 1, while its mask has bit zero clear. Page 9's bit complement represents $-x - 1$, not $-x$. Page 12's addition of 4 is not a bit-2 toggle: $4 + 4 = 8$, whereas XOR toggles exactly the desired bit. On page 14, a half-turn in the 11-bit angle field [51 : 41] requires XOR of word bit 51; the printed mask flips ten bits. Finally, declaring left-to-right notation does not alter C/C++ precedence. For page 16's expression, log-radius 2 and occupancy 0 produce 2 under C rules but 0 under strict left-to-right evaluation. Explicit parentheses or a specified parser are necessary.

Table 1: Source-to-kernel decisions. Page numbers refer to the supplied PDF.

Pages	Construction	Decision and usable scope
7–12	Word fields and Boolean pipeline	Accept explicit widths, masks, signedness and update order; retain all physical numeric bits.
4, 8	Phase sweep	Correct to a declared periodic phase plus cycle count. A sawtooth is not itself a Fourier transform.
5, 9	Hadamard and pinion scaling	The normalized Hadamard is orthogonal. The complete scaled map is not; masked integer code loses information.
5, 11	Golden-ratio rotation	A declared sampling sequence is valid; dyadic shifts are not exact multiplication by ϕ .
3–5, 15	One-bit shape and LUT	Occupancy predicates, not signed distances. Require a domain, geometry and complete LUT.
6–7, 10	Branch culling	Valid only if the bound contains all transformed descendants; the printed proof does not establish this.
6, 12–14	Loop and Klein claim	Finite-state feedback is usable. A parity flip does not prove a Klein-bottle surface or non-orientability.
13–14	Log radius, binary angle	Valid chart and modular arithmetic after domain and mask corrections; quantization is explicit.
16–17	Source OTAN2	Radius and occupancy cannot recover geometric direction. Preserve the actual station-relative vector.
17, 20	Infinite nesting in a word	A 64-bit state has at most 2^{64} distinct values. External state must be accounted for.

Ground-station direction and chart limits. The points (1, 0) and (0, 1) have equal radius and can have equal occupancy, yet their bearings differ by $\pi/2$. The source’s radius/occupancy OTAN2 therefore cannot replace geometric azimuth; it is also distinct from the existing UGTS OTAN2 profile. Keep the physical relative position, ENU axes, range, and defined cases for azimuth/elevation. An 11-bit full-turn angle has a half-bin rounding error of 0.087890625° . At an illustrative $R = 42,164,000$ m,

$$e_{\text{chord}} = 2R \sin(\pi/4096) = 64,678.76 \text{ m.}$$

Even a single-angle 10 m quantization budget at this radius requires at least 24 angular bits; this allocates nothing to other error sources. A station calculation uses actual slant range. Log radius requires $\rho = \log_2(r/r_\star)$ with $r, r_\star > 0$; the origin has no finite value. The page-4 log of a trigonometric product is undefined at zeros and unbounded below, so it cannot establish page 7’s assumed nonnegative monotonic motion. A parent-local interval $[1, 2]$ misses the slit $x = 0$, while its child translated by -1 intersects it. Culling needs an envelope of every descendant. Similarly, $\det \text{diag}(-1, 1, -1) = +1$: two axis flips do not establish non-orientability.

Exact time and derived keys. For integer ticks n , reference tick n_0 and positive integer period P ,

$$n - n_0 = qP + r, \quad q = \left\lfloor \frac{n - n_0}{P} \right\rfloor, \quad 0 \leq r < P.$$

Keeping both q and r preserves time exactly, including negative elapsed ticks; phase alone aliases times separated by a period. Orbital GPST/UTC/TT/TDB conversion and the full epoch remain authoritative. The calendar and constants in the filename derive no orbital initial condition or force constant. Log-polar values, occupancy and phase can supply derived display addresses or keys with declared origin, domain, quantization and winding. They do not replace metre-valued geometry or establish cryptographic unpredictability.

The accepted R10 packing construction. For a 64×64 bit array, the transpose is

$$B'_{c,r} = B_{r,c}, \quad T^2 = I.$$

Six stages exchange corresponding row-index and column-index bits. Each stage is a permutation, so no numeric information is lost. R10 uses this exact transpose for its existing bitplane format, retaining binary64 bit patterns, types, ordering, padding and canonical digest. This is an engineering construction inspired by the source's binary-word organization; the source does not itself specify or prove this algorithm. It imposes no Hadamard arithmetic or golden-ratio quantization on physical coordinates. Release round-trip checks and measured benchmarks establish correctness and speed; this source audit makes no timing claim. Representational exactness does not remove the independently measured physical-model discrepancy.

2 ORBIT-SEED-R1: compact physical seeds and timestamp queries

This profile predicts the position of a satellite relative to a specified ground station. It includes real MEO, GEO, inclined geosynchronous and high elliptical examples. A small, complete model seed is the input; a requested timestamp is the query. The model regenerates position and velocity with explicit dynamics. The two ordered ASA/NA sets and the synchronous JK recurrence select subsequent query times in ordered replay. Direct scrubbing obtains the same physical state without replaying the preceding display interactions.

Complete binary seeds; sizes are measured in the execution table.

Each seed includes the initial state, force constants, compact Sun/Moon and frame coefficients, provenance, both demonstration stations, time/query settings, two ASA/NA mask sets and editable Boolean equations. No future target trajectory table is stored. The native executable and its declared runtime are shared dependencies, accounted for separately from per-seed bytes.

The four fitted states use data at or before 2 January 2025 00:00 GPST. The following seven days are excluded from fitting. Their earlier R1 results were inspected before R2 development, so this interval is a reused benchmark. These are archived examples, not a statement of current satellite positions. An arbitrary future manoeuvre cannot be recovered from a prior seed that does not contain that manoeuvre. Compactness is a consequence of a specified dynamical model and initial conditions; it does not provide lossless compression of every possible orbit.

2.1 The complete state seed

Let the decoded seed be

$$S = (I, t_0, \mathbf{r}_0, \mathbf{v}_0, \mathcal{F}, C, \mathcal{M}, \mathcal{N}, \mathcal{G}, Q, \mathcal{W}, \mathcal{P}), \quad (1)$$

where I identifies the object and model version, $\mathcal{F}$ contains all force constants and fitted accelerations, C contains compact planetary forcing, $\mathcal{M}$ defines the frame/time conversion, $\mathcal{N}$ defines numerical integration, $\mathcal{G}$ defines the stations, Q defines query/chart parameters, $\mathcal{W}$ defines the literal word equations and masks, and $\mathcal{P}$ records construction provenance. Position and velocity are six IEEE 754 binary64 SI numbers, not quantized UGTS addresses.

The upgraded physical model is ORBIT-DYNAMICS-R2; the decoder also retains R1 support. Its state is geocentric and aligned with GCRS/ICRF axes. Full GPST seconds since 6 January 1980 are retained. For an elapsed query τ , $t = t_0 + \tau$ and

$$TT = GPST + 51.184 \text{ s}, \quad JD_{TT} = 2444244.5 + (t + 51.184)/86400. \quad (2)$$

The Chandra input is labelled TDB and is explicitly converted during construction and review. GPST, UTC, TT and TDB calendar labels are not interchanged. At this example epoch GPST is 18 seconds ahead of UTC.

2.2 Explicit Earth frame and compact forcing

Define the passive Earth rotation matrix and polar-motion matrix as

$$R_z(a) = \begin{pmatrix} \cos a & \sin a & 0 \\ -\sin a & \cos a & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (3)$$

$$P = \begin{pmatrix} c_x & 0 & s_x \\ s_y s_x & c_y & -s_y c_x \\ -c_y s_x & s_y & c_y c_x \end{pmatrix}, \quad c_x = \cos x_p, \quad s_x = \sin x_p, \quad c_y = \cos y_p, \quad s_y = \sin y_p. \quad (4)$$

With the embedded slowly varying celestial matrix $Q(\tau)$,

$$a(\tau) = a_0 + \dot{a}\tau + \delta a(\tau), \quad M(\tau) = P(\tau)R_z(a(\tau))Q(\tau), \quad (5)$$

$$\mathbf{r}_E = M\mathbf{r}, \quad \mathbf{v}_E = M\mathbf{v} + \dot{M}\mathbf{r}. \quad (6)$$

Thus Earth-fixed velocity includes the time derivative of the frame. The construction uses IAU 2006/2000A and the IERS Bulletin A published on 26 December 2024, before the forecast epoch. Published daily x_p, y_p and DUT1 values are linearly interpolated. The ERA correction is $\delta a(\tau) = \dot{a}[\text{DUT1}(\tau) - \text{DUT1}(0)]$, with $\dot{a} = 2\pi(1.00273781191135448)/86400$. The velocity derivative includes $\dot{P}$, $\dot{\delta a}$ and $\dot{Q}$. TIO locator s' is zero. Published EOP predictions carry their own uncertainty; their interpolation is not a future observed-EOP measurement.

Each slowly varying scalar coefficient block uses a finite Chebyshev expansion:

$$u(\tau) = \frac{2\tau - \tau_a - \tau_b}{\tau_b - \tau_a}, \quad f(\tau) = \sum_{k=0}^d c_k T_k(u), \quad T_0 = 1, \quad T_1 = u, \quad T_{k+1} = 2uT_k - T_{k-1}. \quad (7)$$

The seed stores the interval, degree and every coefficient. Q has nine degree-eighteen rows; Sun and Moon forcing have three rows of degree sixteen and thirty respectively over the fifteen-day construction domain. Daily EOP blocks have three degree-one rows: $\delta a, x_p, y_p$. Rapid Earth rotation remains a direct trigonometric rotation. At shared segment boundaries the later segment applies. Queries outside the embedded domain return an error rather than silently extrapolating the coefficient blocks.

The Sun/Moon coefficients approximate the DE440s planetary model published in 2020. They describe known forcing from other bodies, not withheld positions of the target satellite. The measured off-node forcing discrepancies are about 1.99 m for the Sun and 0.060 m for the Moon. These quantities are not satellite orbit accuracy. Original DE440 files are unnecessary during packed replay.

2.3 Actual orbital differential equations

The physical state evolves as

$$\frac{d\mathbf{r}}{d\tau} = \mathbf{v}, \quad \frac{d\mathbf{v}}{d\tau} = M^T \nabla_{\mathbf{p}} U(\mathbf{p}) + \mathbf{a}_{\odot} + \mathbf{a}_{\text{Moon}} + \mathbf{a}_{\text{SRP}} + \mathbf{a}_{\text{RTN}} + \mathbf{a}_{\text{GR}}, \quad \mathbf{p} = M\mathbf{r}. \quad (8)$$

Here the Moon acceleration symbol denotes the declared lunar third-body term. For $R = \|\mathbf{p}\|$, latitude sine $s = p_z/R$, longitude λ , Earth radius a_E and parameter μ , the full degree/order twelve

field is

$$U(\mathbf{p}) = \frac{\mu}{R} \sum_{n=0}^{12} \left(\frac{a_E}{R} \right)^n \sum_{m=0}^n P_{nm}(s) [C_{nm} \cos(m\lambda) + S_{nm} \sin(m\lambda)]. \quad (9)$$

P_{nm} has no Condon–Shortley phase. The seed stores unnormalized C_{nm} , S_{nm} , converted from the EGM96 fully normalized coefficients by

$$C_{nm} = \sqrt{(2 - \delta_{m0})(2n + 1) \frac{(n - m)!}{(n + m)!}} \bar{C}_{nm}, \quad (10)$$

and the same factor for S . It includes $C_{00} = 1$ and $S_{00} = 0$. The positive-potential convention uses $+\nabla U$.

Native evaluation avoids longitude and polar division. Define $X = a_E p_x / R^2$, $Y = a_E p_y / R^2$, $Z = a_E p_z / R^2$, $D = a_E^2 / R^2$, $V_{00} = a_E / R$ and $W_{00} = 0$. The Cartesian Cunningham recurrence is

$$V_{mm} = (2m - 1)(XV_{m-1,m-1} - YW_{m-1,m-1}), \quad (11)$$

$$W_{mm} = (2m - 1)(XW_{m-1,m-1} + YV_{m-1,m-1}), \quad (12)$$

$$V_{nm} = \frac{2n - 1}{n - m} ZV_{n-1,m} - \frac{n + m - 1}{n - m} DV_{n-2,m}, \quad (13)$$

$$W_{nm} = \frac{2n - 1}{n - m} ZW_{n-1,m} - \frac{n + m - 1}{n - m} DW_{n-2,m}, \quad (14)$$

$$U = \frac{\mu}{a_E} \sum_{n=0}^{12} \sum_{m=0}^n (C_{nm} V_{nm} + S_{nm} W_{nm}). \quad (15)$$

The diagonal recurrence uses $m = 1, \dots, 12$; the off-diagonal recurrence uses $m = 0, \dots, 12$ and $n = m + 1, \dots, 12$. For $n = m + 1$ the second term is absent. Each native recurrence value carries three analytic Cartesian derivatives; the product rule produces ∇U . Independent Python validation differentiates the scalar potential with complex steps, and native tests compare a separately coded spherical-gradient oracle. The constants are $\mu = 3.986004415 \times 10^{14}$ SI and $a_E = 6378136.3$ m. This is the tide-free EGM96 field truncated at degree/order twelve.

For each third body b with Earth-centred position $\mathbf{r}_b$,

$$\mathbf{a}_b = \mu_b \left[\frac{\mathbf{r}_b - \mathbf{r}}{\|\mathbf{r}_b - \mathbf{r}\|^3} - \frac{\mathbf{r}_b}{\|\mathbf{r}_b\|^3} \right]. \quad (16)$$

The indirect term removes the acceleration of the Earth. With $\mathbf{d} = \mathbf{r} - \mathbf{r}_\odot$, the fitted radiation-pressure term is

$$\mathbf{a}_{SRP} = \chi A \left(\frac{\text{AU}}{\|\mathbf{d}\|} \right)^2 \frac{\mathbf{d}}{\|\mathbf{d}\|}. \quad (17)$$

$\chi = 0$ in the declared cylindrical Earth shadow when the seed's shadow flag is enabled, and 1 otherwise. With shadow disabled, $\chi = 1$ everywhere. Explicitly, if $\hat{\mathbf{s}} = \mathbf{r}_\odot / \|\mathbf{r}_\odot\|$, shadow requires $\mathbf{r} \cdot \hat{\mathbf{s}} < 0$ and $\|\mathbf{r} - (\mathbf{r} \cdot \hat{\mathbf{s}})\hat{\mathbf{s}}\| < a_E$. There is no hidden penumbra, attitude or manoeuvre model. Finally,

$$\hat{\mathbf{R}} = \mathbf{r} / \|\mathbf{r}\|, \quad \hat{\mathbf{N}} = (\mathbf{r} \times \mathbf{v}) / \|\mathbf{r} \times \mathbf{v}\|, \quad \hat{\mathbf{T}} = \hat{\mathbf{N}} \times \hat{\mathbf{R}}, \quad (18)$$

$$\mathbf{a}_{RTN} = a_R \hat{\mathbf{R}} + a_T \hat{\mathbf{T}} + a_N \hat{\mathbf{N}}. \quad (19)$$

The Schwarzschild correction is explicit, with $r = \|\mathbf{r}\|$:

$$\mathbf{a}_{GR} = \frac{\mu}{c^2 r^3} \left[\left(\frac{4\mu}{r} - \mathbf{v} \cdot \mathbf{v} \right) \mathbf{r} + 4(\mathbf{r} \cdot \mathbf{v})\mathbf{v} \right], \quad c = 299792458 \text{ m/s}. \quad (20)$$

The three signed empirical accelerations and A are fitted only on the declared training arc. Their continuation is a physical approximation, not knowledge of future spacecraft operations. The same ten-parameter fit is fixed for all four objects: state six, SRP one and RTN three. Earlier training-window candidates of one, three and seven days are compared against the earlier January 1 interval; the lowest maximum error selects the window. The final fit uses only data at or before the January 2 cutoff. This past-only selection is distinct from the already inspected January 2–9 comparison benchmark. Harmonics above degree twelve, tides, detailed radiation pressure, attitude changes and manoeuvres remain possible sources of forecast disagreement.

2.4 Bottom-up timestamp reconstruction and scrubbing

Native CPU/CUDA integrate these equations with classical RK4. For a six-vector $\mathbf{y} = (\mathbf{r}, \mathbf{v})$ and derivative F ,

$$\mathbf{k}_1 = F(t, \mathbf{y}), \quad \mathbf{k}_2 = F(t + h/2, \mathbf{y} + h\mathbf{k}_1/2), \quad (21)$$

$$\mathbf{k}_3 = F(t + h/2, \mathbf{y} + h\mathbf{k}_2/2), \quad \mathbf{k}_4 = F(t + h, \mathbf{y} + h\mathbf{k}_3), \quad (22)$$

$$\mathbf{y}^+ = \mathbf{y} + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (23)$$

MEO/GEO/IGSO use $h = 30$ seconds; Chandra uses 7.5 seconds, selected through an independent numerical-convergence check. Its fitted initial state and forces were not refitted against future positions. Independent Python propagation uses DOP853 rather than the native RK4 implementation.

For any elapsed time τ , define $n = \text{trunc}(\tau/h)$ and $\epsilon = \tau - nh$. Canonical lattice states are obtained by repeated signed steps from the packed epoch. The exact requested state is one final fractional step of length ϵ from the canonical state at nh . Fractional results never become canonical checkpoints. This makes the physical query independent of the order in which the timeline was scrubbed.

The CPU worker retains at most the declared number of sparse checkpoints (256 in the examples, at one-hour spacing). Missing checkpoints regenerate from an available earlier canonical state or the epoch. Eviction changes work, not the mathematical trajectory. Cold-query work is proportional to elapsed integration steps; compact storage does not imply constant-time propagation. CUDA batches assign independent timestamp queries to device work and integrate from the epoch. They currently do not share the CPU checkpoint cache. CPU scrubbing and parallel CUDA batch execution are distinct measured paths.

2.5 Two vacuum/set stages and synchronous JK

The user's "double vacuum, double set" is concretely defined here as exactly two independently stored ASA/NA/boundary mask triples, (A_1, N_1, B_1) and (A_2, N_2, B_2) , applied in order. No unspecified operation is inferred from the phrase. All operators below are 32-bit word operators, and P is the present mask. For stage i and input w ,

$$a_i(w) = w \& A_i \& P, \quad n_i(w) = a_i(w) \& N_i, \quad (24)$$

$$h_i(w) = \text{popcount}(n_i(w) \& B_i), \quad V_i(w) = \begin{cases} 0, & h_i(w) > 0, \\ n_i(w), & h_i(w) = 0. \end{cases} \quad (25)$$

One selected boundary hit therefore absorbs the entire output word. It does not merely clear the intersecting bit. At ordered query j ,

$$\mathbf{s}_j = \Phi(\mathcal{S}, \tau_j), \quad g_j = G(\mathbf{s}_j, \mathcal{G}, \tau_j), \quad (26)$$

$$d_j = E_D(g_j, \tau_j), \quad x_j = E_X(q_j, d_j), \quad (27)$$

$$y_j = V_2(V_1(x_j)), \quad J_j = E_J(q_j, y_j, d_j), \quad (28)$$

$$K_j = E_K(q_j, y_j, d_j), \quad q_{j+1} = [(J_j \& \neg q_j) \mid (\neg K_j \& q_j)] \& P, \quad (29)$$

$$\tau_{j+1} = \min(\tau_{end}, \tau_j + \Delta(q_{j+1})). \quad (30)$$

Both JK right-hand sides read the same old q_j . Boundary absorption does not implicitly freeze JK. The editable X expression accepts q, d ; J and K accept q, y, d . They support word constants zero/ones and bitwise AND, OR, XOR and complement with parentheses. They compile to literal Boolean truth tables; an independent AST interpreter checks the complete trace.

Default predicate bits are, in ascending order: above the station elevation mask, within three degrees of that mask, absolute seed age $|\tau|$ at least 24 hours, positive elevation rate, and approaching range. The actual default expressions are

$$x = q \mid d, \quad J = y \& d, \quad K = \neg d. \quad (31)$$

With both unabsorbed identity mask sets these reduce to $q^+ = d$. The state still selects the next physical query timestamp: $\Delta = 30$ seconds when $(q^+ \& 6) \neq 0$, and 300 seconds otherwise. Different mask/equation inputs change that recurrence, as the counterfactual and absorption tests demonstrate. Physical acceleration never reads q ; the CGK spring equations are retained separately rather than renamed orbital gravity.

2.6 Station-relative geometry and both horizons

For WGS84 station position $\mathbf{r}_g$ and its standard East/North/Up rotation $L(\varphi, \lambda)$,

$$\mathbf{z} = (E, N, U)^T = L(\mathbf{r}_E - \mathbf{r}_g), \quad \dot{\mathbf{z}} = L\mathbf{v}_E. \quad (32)$$

Set $H = \sqrt{E^2 + N^2}$ and $D = \|\mathbf{z}\|$. The returned quantities are

$$az = \text{atan2}(E, N) \bmod 2\pi, \quad el = \text{atan2}(U, H), \quad (33)$$

$$\dot{D} = \mathbf{z} \cdot \dot{\mathbf{z}} / D, \quad \frac{d \sin el}{dt} = \dot{U}/D - U\dot{D}/D^2. \quad (34)$$

Azimuth is clockwise from North. At the horizontal origin azimuth and its derivative are explicitly undefined. Angles are simultaneous geometric values: there is no atmospheric refraction or retarded light-time correction here.

Visibility events solve

$$f(t) = \sin el(t) - \sin el_{mask} = 0. \quad (35)$$

The search refines sign brackets and derivative-bracketed extrema, so it can recover a short pass whose sampled endpoints are both below the mask. It reports grazing candidates and numerical root tolerances. The validation compares 120-second and 30-second searches over seven days for both stations and every object. Without a global interval bound this is not a proof that arbitrarily many roots cannot occur between samples. Always-visible/hidden labels mean visible/hidden over the searched interval under the stated search policy.

The second horizon is an error-budget horizon. Given withheld reference samples $\mathbf{r}_k^*$, $e_k = \|\mathbf{r}(t_k) - \mathbf{r}_k^*\|$. For budget b the report records the first sample with $e_k > b$ and its predecessor. The interval brackets a sampled exceedance; it does not certify continuous error below b before it. An epoch can be inside the model's coefficient domain while outside a useful accuracy horizon. Maximum error across the whole interval is reported alongside endpoint error because an orbit can diverge and later approach the reference.

2.7 Original UGTS keys and complete time

The orbital chart uses $\rho = \log(H/r_0)$, $\theta = \text{atan2}(N, E)$ and the explicit separate hoop angle ϕ . The source chart is used only when $-20 \leq \rho \leq 0$; out-of-range coordinates are not wrapped into a valid key. The full tick is $k = \lfloor t/\delta t \rfloor$, with integer winding and remainder from $k = 16384w + k_{phase}$. The normalized fraction is separately labelled $k_{phase}/16384$. Source quantization produces the original (20, 18, 14, 12)-bit tuple. Both the contiguous and original Morton layouts are retained. Full position, velocity, Up, absolute time and provenance remain in the result. Source OTAN2 uses $\Delta\theta = \text{wrap}(\theta_j - \theta_{j-1})$ and is still $\arctan(\Delta\theta/\Delta\rho)$, with its explicit undefined-ratio cases, and is not replaced by azimuth.

2.8 Binary packing, integrity and executable entry points

The 52-byte little-endian header contains magic ATOMORB1, format/codec versions, compressed and raw lengths, and the SHA-256 of the domain-prefixed canonical payload. The payload uses tagged null/Boolean/signed and unsigned 64-bit integer/binary64/string/list/map values. Map keys are sorted, unique UTF-8 strings. Codec 1 compresses those canonical bytes directly. The default codec 2 packs the numeric bit patterns into one-bit planes before Zlib compression. It preserves all 64 bits per numeric value. For numeric word w_j , define

$$P_{b,g} = \sum_{\ell=0}^{63} [(w_{64g+\ell} \text{ shr } b) \& 1] 2^\ell, \quad b = 0, \dots, 63. \quad (36)$$

Absent final lanes are zero. The inverse restores bit b of word j from lane $j \bmod 64$ of $P_{b,\lfloor j/64 \rfloor}$. The plane-major uint64 words follow a structural template with original numeric type tags, and a 16-byte BITPLN64 header with template length and word count. Signed integers, unsigned integers and binary64 preserve their original bits; the structural template retains Boolean tokens. Both codecs share the same canonical content digest. Plane packing adds zero numerical error, but does not change floating-point arithmetic in propagation. The measured example packs are slightly larger with this representation than with codec 1. The decoder checks exact lengths, bounded expansion and nesting, complete stream termination, canonical re-encoding, digest, model schema and time domains. There is no executable payload or pickle.

The digest domain is atomOS:ORBIT-SEED-R1:canonical-binary followed by a zero byte. Physical-model and complete-seed hashes differ by scope. Runtime timings and cache occupancy are excluded from the deterministic trace digest; all mathematical query fields and both mask/JK stages remain covered. Independent verification reconstructs the native orbit, checks the full station geometry, keys/time, OTAN2, the AST recurrence, next query time and final chain.

```
python tools/orbital_seed.py inspect --seed examples/orbit/seeds/C03.orbseed
python tools/orbital_seed.py query --seed examples/orbit/seeds/C03.orbseed --worker bin
    /cpu/orbit_worker.exe --station singapore_demo --time-s 12345.678 --out NEW_QUERY.
    json
python tools/orbital_seed.py predict --seed examples/orbit/seeds/G05.orbseed --worker
    bin/cpu/orbit_worker.exe --end-s 86400 --out NEW_RUN
python tools/verify_orbit_run.py --seed examples/orbit/seeds/G05.orbseed --worker bin/
    cpu/orbit_worker.exe --run NEW_RUN --out NEW_VERIFY.json
python tools/orbit_scrub.py --worker bin/cpu/orbit_worker.exe
```

The local timeline is served at <http://127.0.0.1:3619>. It accepts exact fractional-second queries, a slider, object/station selection, ordered feedback playback and visibility events. The JSON companions are editable; packing them rebuilds the binary .orbseed files with a new content identity. Accuracy reports are associated only when their canonical physical-model hash matches the loaded seed.

2.9 Packed execution and regression evidence

The following numbers are derived from the final decoded seeds and verified native binaries. The isolated replay child receives only the packed seed, shared runtime source and native executable; an audit hook denies every read under the original release directory. The reconstructed fractional, past and seven-day timestamp states match the normal native run bit for bit.

Object	Seed bytes	Default queries	Altered JK	Warm max (ms)
G05	9,066	397	2,881	0.816
C03	9,074	289	2,881	0.956
C06	9,073	417	2,881	0.834
CHANDRA	8,716	520	2,881	2.075

Query counts cover one day. The counterfactual sets $J = 1, K = 0$ and changes the actual timestamps selected by the native recurrence. Physical states at shared timestamps remain identical. Warm timing is end-to-end Python/native geometry latency on this host for the recorded fractional seek sequence, not a universal performance bound. Cold seeks may require thousands of integration steps.

Across eight seven-day station/object searches, 83 rise/set events were found. The largest event-time change under refinement from 120-second to 30-second sampling was 0.002508 seconds. The event algorithm's grazing and no-global-root-exclusion qualifications still apply.

A five-minute, seven-day table containing time and six binary64 state components would occupy 112,952 bytes before its metadata. This compares a generated state table to its specified model, not arbitrary-trajectory compression. The demonstration archive also contains large reference datasets and audit traces, which are not per-seed replay inputs.

The table is 12.45 times the size of the largest complete packed seed. The shared CPU executable is 77,824 bytes, and the isolated shared Python source is 89,342 bytes at validation. Installed libraries and the OS/C++ runtime are additional shared dependencies. The default codec packs 64 one-bit planes into uint64 words without discarding any numeric bit. The canonical codec remains readable; bit-plane packing is slightly larger for these examples.

The new native orbital checks compare 1,040 CPU/CUDA positions across the four models, with maximum disagreement 0.00004320 m. All 256 complete CPU/CUDA two-stage word traces match. Bounded-cache stress tests remain bit-exact through repeated eviction. Compute Sanitizer reports zero errors for 129 orbital queries and the word operation. The full Python suite passes 139 tests, including bit-plane reconstruction, rehashed-false-geometry, OTAN2 rejection and missing-reference-interval acceptance.

Fresh retained-component regression passes four CTests on each build, 256 SATNAV epochs per backend with independent reconstruction, 4,225 SRK transitions and 3,308 CGK transitions per backend, and 6,861 persistent live-worker comparisons per backend. The required GPU review tool also records zero errors under both texture and global-memory Compute Sanitizer runs on the actual RTX 5070 Ti Laptop.

The first orbital RK4 test exposed an omitted final state assignment; that defect was fixed before these final results. Earlier failed and superseded investigation folders are retained as such and are not used as final validation. Numerical agreement is separate from the physical future-reference errors below.

2.10 R2 physical-position results and separate confirmation

The R2 models stay below the adopted **10 m three-dimensional reference-discrepancy budget at every tested sample through 24 hours** for all four targets, on both the January benchmark and the separate February confirmation date. This statement concerns sampled positions of these seeds; it is not a universal or continuous-time orbit guarantee. The January window was previously inspected during R1 development and is explicitly a **reused benchmark**. The February selection procedure was frozen before its future errors were evaluated; no model was retuned afterward.

The forecast quantity is

$$e_i = \|\hat{\mathbf{r}}(t_i) - \mathbf{r}_{\text{reference}}(t_i)\|_2, \quad t_i > t_0.$$

The January origin is 2 January 2025 00:00 GPST, $t_0 = 1419811200$; February uses 2 February 2025 00:00 GPST, $t_0 = 1422489600$. All cutoff and earlier target rows are excluded from the forecast curves. R2 selects among 1-, 3- and 7-day training windows using an internal validation day entirely before the forecast origin. The selected January windows are 3/3/3/1 days for G05/C03/C06/Chandra; the unchanged rule selects 1/7/3/3 days for February. External EGM96 gravity and planetary forcing are distinct from future target observations.

Table 2: Maximum discrepancy over all first-day samples (metres). Each R2 column has 288 samples per target through 24 h; Chandra’s final sample is approximately 51.185 s before the exact endpoint.

Target	R1 January	R2 January	R2 February
G05 / MEO	285.330	1.057	3.072
C03 / GEO	20.600	5.304	4.654
C06 / IGSO	44.684	1.172	2.108
Chandra / HEO	76.506	8.093	0.290

Table 3: R2 January discrepancy in metres at the nearest actual sample to each requested horizon. Individual samples are not worst-case interval errors.

Target	15 min	1 h	6 h	24 h	72 h	7 d
G05 / MEO	0.474	0.655	0.272	0.713	2.228	12.531
C03 / GEO	1.368	1.702	4.350	4.172	17.939	73.481
C06 / IGSO	0.547	0.609	0.615	1.172	2.335	2.360
Chandra / HEO	0.003	0.008	0.074	2.052	28.338	3680.247

Time grid. GNSS is on the exact requested GPST grid. Chandra is supplied on a 300 s TDB grid, converted using the periodic TDB–TT correction and $\text{GPST} = \text{TT} - 51.184 \text{ s}$. The nearest 15 min sample is 848.816057 s after the January origin and 848.815204 s after the February origin. The final samples are respectively 604748.815856 s and 604748.815036 s. Exact offsets remain in the CSV/JSON, including all six requested horizons for both dates.

Table 4: R2 whole-window statistics (metres). January has 2,016 samples per target. February has 2,016 each except C03, which has 1,441 and a 48 h gap.

Target	Jan. RMS	Jan. maximum	Feb. RMS	Feb. maximum
G05 / MEO	5.294	12.998	33.113	71.800
C03 / GEO	37.434	78.717	39382.862	100604.174
C06 / IGSO	2.943	8.084	3.494	7.629
Chandra / HEO	5412.235	25635.951	2055.685	5663.182

The seven-day 10 m budget is **not met for all targets**. C06 stays below it at every seven-day sample on both dates; G05 exceeds it later on both dates. C03 and Chandra have additional reference qualifications below. The original R1 seven-day maxima were 7402.960/532.582/1237.242/19399.307 m in the same object order. The R2 first-day improvement must not be extended into a claim that every seven-day maximum improved or passed. RMS and percentiles describe correlated samples, not independent-sample confidence intervals. Reports additionally retain interval RMS/maxima and sample percentiles.

2.11 Sampled thresholds and incomplete reference coverage

Table 5: Preceding sample and first sampled exceedance, in hours after the origin, rounded to 0.001 h. These pairs are not certified continuous crossing brackets. “None” means no exceedance at the available samples.

Date	Target	10 m	100 m	1 km
January	G05 / MEO	(145.083, 145.167]	None at samples	None at samples
January	C03 / GEO	(29.000, 29.083]	None at samples	None at samples
January	C06 / IGSO	None at samples	None at samples	None at samples
January	Chandra / HEO	(50.402, 50.486]	(82.569, 82.652]	(83.986, 84.069]
February	G05 / MEO	(50.833, 50.917]	None at samples	None at samples
February	C03 / GEO	(72.000, 120.000]	(72.000, 120.000]	(72.000, 120.000]
February	C06 / IGSO	None at samples	None at samples	None at samples
February	Chandra / HEO	(83.986, 84.069]	(83.986, 84.069]	(83.986, 84.069]

For 10 km, January Chandra first exceeds the level at 302648.815956 s, following the sample at 302348.815956 s; all January GNSS targets remain below 10 km at their samples. February C03 first exceeds 10 km at the first sample after its gap, 432000 s; the previous sample is at 259200 s. Other February targets remain below 10 km at the sampled times.

C03 February coverage. GFZ contains no C03 records on February 5 and 6. Adjacent daily endpoints leave a 48-hour reference gap and 575 missing five-minute samples. Wuhan also lacks C03 on February 6 and cannot certify the missing interval. GFZ remains the predeclared primary source. The first post-gap discrepancy is 29256.547 m and the later maximum is 100604.174 m; these observed exceedances are retained. Full-window coverage is **indeterminate**, and no continuous tolerance passage through the gap is inferred. No maneuver or other cause is assigned from missing records alone. The figures leave the missing interval blank.

2.12 Numerical error and reference discontinuities

Table 6: Separate numerical and frame comparisons, in metres. January’s independent physical audit uses 11 times across the past/future domain; February compares nine future checkpoints with adaptive DOP853. The frame column is the larger of the two complete sampled frame-approximation maxima.

Target	Jan. RK4/DOP853	Feb. RK4/DOP853	Frame approximation
G05 / MEO	0.031716	0.026461	0.002007
C03 / GEO	0.001232	0.001023	0.003260
C06 / IGSO	0.001252	0.001026	0.003169
Chandra / HEO	0.036582	0.147044	0.009135

Native RK4 uses 30 s for GNSS and 7.5 s for Chandra. Same-seed integration agreement tests arithmetic, not orbit truth. The January forecast report’s own DOP853 check covers only its

first sample; the wider January evidence above comes from the separate, target-independent physical audit. Native CPU/CUDA checks and exact word/packing checks remain separate evidence. The frame test compares the embedded daily-EOP/Chebyshev frame with direct SOFA IAU 2006/2000A evaluation of the same published prediction. It excludes actual future EOP error, station uncertainty and a complete relativistic ICRF/GCRS coordinate error budget; it is not subtracted as an independent scalar variance.

Chandra source discontinuities. The January error rises from 123.204 m to 23839.033 m between two five-minute samples. A separate one-second Horizons query shows that at January 5 TDB 12:01:09–12:01:10 the source position displacement differs from the integral of its own reported velocity by **24317.077 m**; outside the neighboring jump intervals the same local check is at most 0.0654 m. Native RK4 agrees with independently tightened DOP853 to about 1.5 mm at the two forecast endpoints surrounding that jump. The source header identifies CFA’s merged trajectory, but no local join or maneuver cause. The large discontinuity therefore cannot be attributed solely to smooth force error or native integration.

The February reference has a corresponding kinematic discontinuity at February 5 TDB 12:01:09–12:01:10: **2487.123 m** displacement-minus-velocity integral mismatch, versus at most 2.26 μm away from its neighboring intervals. The forecast’s next five-minute error rises from 1.053 m to 2492.098 m. These diagnostics use only the source’s positions and velocities; no model is fitted to them. **All original discrepancy samples and whole-window maxima are retained.** The cause of the provider discontinuities is not established, and no corrected “true orbit” accuracy is invented. Original queries, hashes and calculations are in `source/orbit_data/diagnostics/`.

Reference uncertainty. January GFZ/Wuhan comparisons give 3D RMS 0.0263 m for G05, 2.074 m for C03 and 0.1692 m for C06; C03’s maximum provider disagreement is 3.575 m. These are product differences, not certified absolute bounds. Providers share some underlying observations. Chandra supplies no state covariance or absolute position guarantee. A millimetre or sub-metre match to a selected reference is therefore not a matching physical-accuracy claim.

2.13 Chronology, source conventions and reproducible evidence

The R2 physical model uses the documented degree/order-12 EGM96 field, cutoff-available daily Earth orientation, and external Sun/Moon forcing. EGM96’s original μ , reference radius, tide-free convention and fully normalized coefficients are preserved with explicit conversion in `source/orbit_data/gravity_reference/`. The format’s normalization default is documented by ICGEM: <https://icgem.gfz.de/docs/ICGEM-Format-2023.pdf>.

The EOP sources were published before their cutoffs: IERS Bulletin A of 26 December 2024 and 30 January 2025, respectively: <https://datacenter.iers.org/data/6/bulletina-xxxvii-052.txt> and <https://datacenter.iers.org/data/6/bulletina-xxxviii-005.txt>. Both permit public release and unlimited distribution. The February GFZ reference changes from IGS20 to IGB20. IGSMail-8543, published 9 December 2024, specifies zero datum transformation parameters because origin, scale and orientation remain aligned; individual reference-station coordinates are updated. Preserve each source label and apply no fitted alignment: <https://lists.igs.org/pipermail/igsmail/2024/008539.html>.

Precise target products were retrieved retrospectively; their earlier estimates can incorporate later observations within provider processing. Neither date proves that identical seeds were available live in 2025. January is a reused comparison, while February applies the already-frozen selection rule to a separate date. The confirmation freeze manifest binds the rule and model-file hashes. Raw source acquisition and coverage inspection preceded any February prediction-error evaluation. No later error was used to refit a seed.

- **GFZ and IGS.** Retain GFZ/IGS provider and network attribution. The cited GFZ product-series DOI is <https://doi.org/10.5880/GFZ.1.1.2016.003>; its DataCite record declares

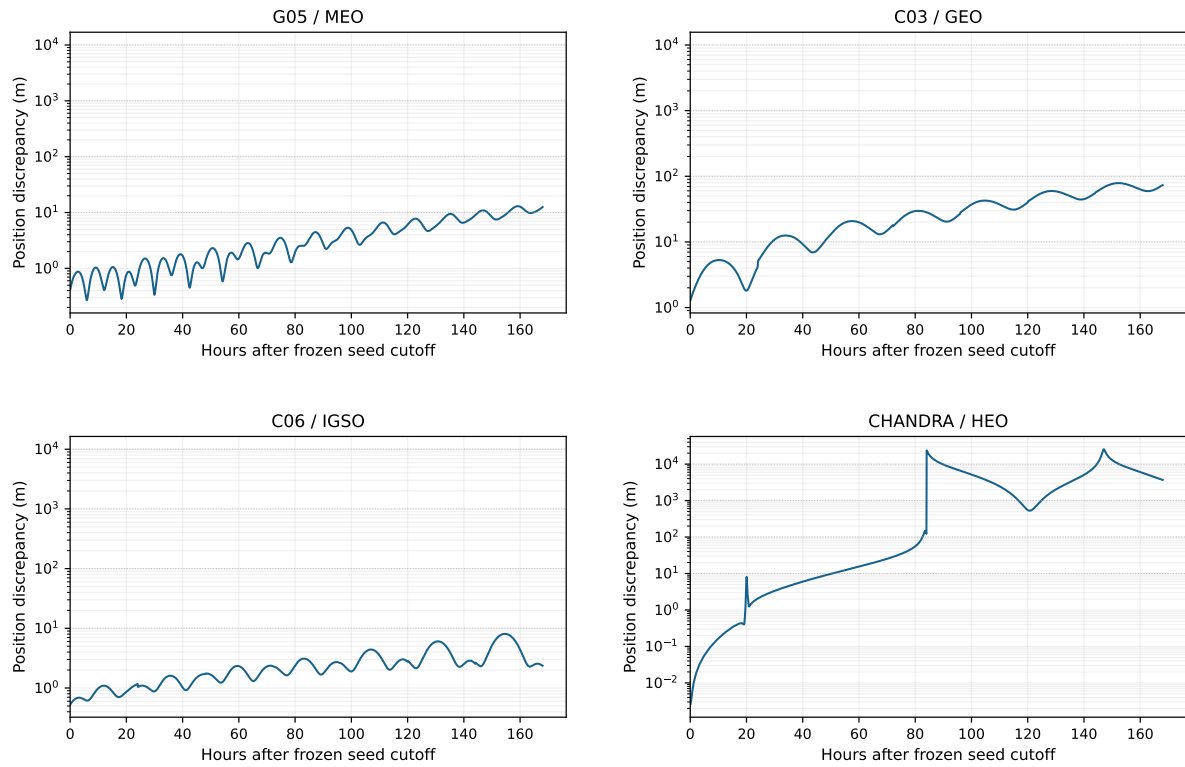
CC BY-NC 4.0. The DOI names an ultra-rapid series whereas these GBM files are rapid products without a separate per-file grant. The fixture bundle conservatively retains that noncommercial qualification and is not claimed as unrestricted commercial data. IGS terms: <https://www.igs.org/wp-content/uploads/2020/09/IGS-Data-and-Product-Disclaimer-and-Terms-of-Use-200805.pdf>.

- **CODE and Wuhan.** CODE's reference <https://doi.org/10.48350/197028> lists open access/BORIS terms, not a blanket MIT grant. Wuhan final products retain provider/IGS attribution. Original archive URLs and per-file hashes remain in the source manifests.
- **NASA/JPL and CFA.** Acknowledge Giorgini and the JPL Solar System Dynamics Group, NASA/JPL Horizons, and CFA for Chandra: <https://ssd.jpl.nasa.gov/horizons/>. Retrieval date is 15 September 2026. Geometric ICRF/TDB settings and original responses are retained. Sun/Moon Horizons responses identify DE441; the construction oracle is DE440s, a distinct planetary ephemeris version.

Authoritative reports. Use `results/orbit_accuracy_r2_cpu_verified/accuracy_summary.json`, its CUDA counterpart, and `results/orbit_accuracy_confirmation_cpu_verified/accuracy_summary.json`. R1 evidence remains in `results/orbit_accuracy_cpu_final/` for comparison. The pre-fix failed trial and superseded intermediate models are not final accuracy evidence. CSVs retain every measured error, exact horizon offset, source and frame. Each report distinguishes `model_file_sha256` from `physical_model_sha256=orbit_seed.digest(model)`; legacy `model_sha256` is explicitly the file-hash alias. Accuracy displayed for a custom packed seed applies only when the physical-model digest matches. Reference fixtures, publications and the DE440s construction binary are separate from the complete replay seed byte count.

2.14 R2 January benchmark curves

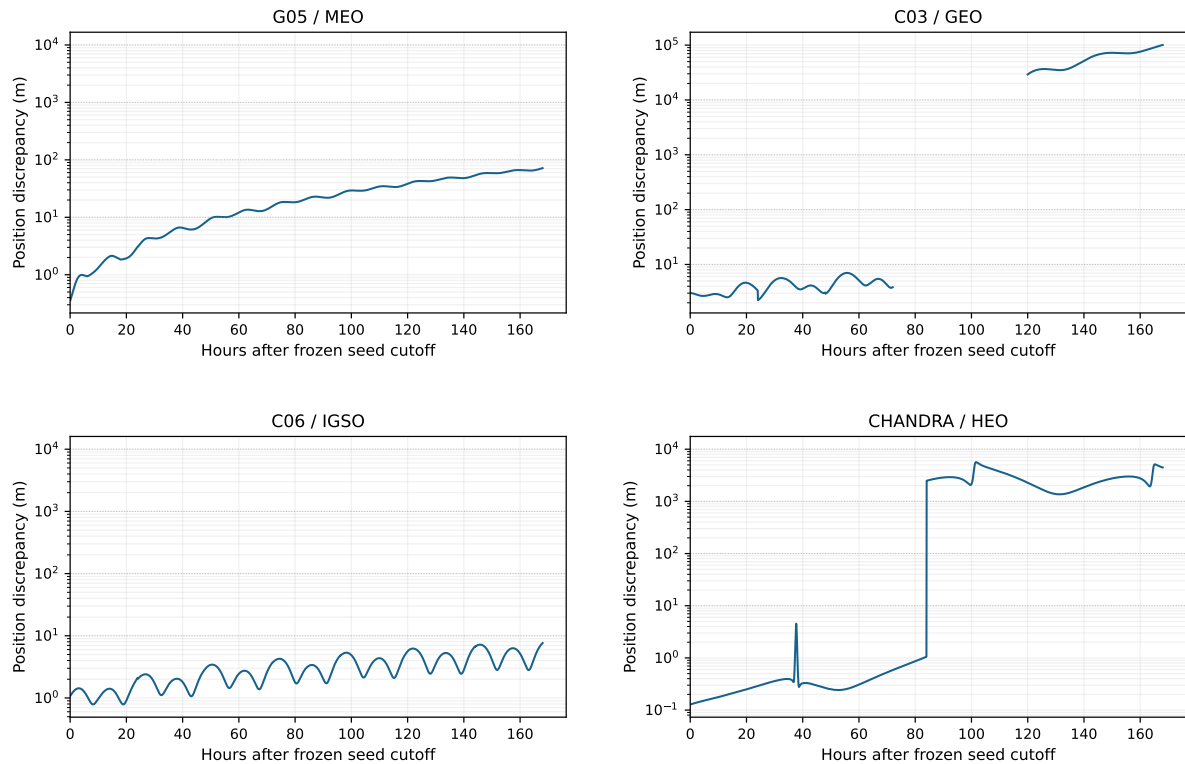
Frozen-seed forecast versus later external trajectory samples



This is the reused January benchmark. Dotted lines are reporting thresholds, not fitted constraints. The Chandra source discontinuity remains visible and its errors remain counted. First-day detail is also supplied as `forecast_error_first_day.pdf` in the same result directory.

2.15 Separate February confirmation curves

Frozen-seed forecast versus later external trajectory samples



The selection rule was frozen before this date's errors were evaluated. The C03 reference gap is blank; connecting its endpoints would imply unavailable measurements. Chandra's source discontinuity is retained. Lines join only available adjacent samples and do not certify error between samples or beyond the declared window.

Retained receiver, geometry and source contracts

The following sections retain the earlier executable mathematical profiles and their dated measurements. References to the 3.6.1.8 live receiver are historical. The current orbital seed profile, timestamp queries and satellite forecast evidence are in Section 2. The baseline regression results bind the preserved component binaries; the optimization chapter reports fresh checks of this release without relabelling the earlier receiver measurements as orbital accuracy.

3 Live GPS receiver-to-position pipeline

LIVE-GPS-L1-R1 supplies the actual GPS equations and transport inputs missing from the prepared-observation kernel. The implemented chain is

$$\text{RTCM bytes} \longrightarrow (P_i, e_i, t_r) \longrightarrow (x_{s,i}^{rx}, \delta t_{s,i}, I_i, T_i) \longrightarrow (x_r, y_r, z_r, b_r). \quad (37)$$

The raw code P_i and satellite ephemeris e_i come from a real remote receiver; the physical corrections are calculated locally. The orbit and atmospheric model is Python; the position/clock solution is the existing FP64 native CPU/CUDA kernel, reached through a persistent process. This closes the reception-to-position chain without assuming an upstream program has already solved the geometry.

Meaning of the live demonstration

The user has no local GNSS hardware. The Internet supplies observations from the LIENSS reference receiver in the public Centipede network. Every computed coordinate in this experiment locates that remote receiver. Its transmitted antenna reference point (ARP) is held out of the estimator and used only for subsequent comparison. No host-computer location is inferred from it.

3.1 What crosses the network boundary

The bounded client performs a read-only NTRIP/HTTP GET, or reads a TCP byte stream. It supports HTTP chunk framing, optional ICY headers, timeouts and bounded reconnection. Centipede permits one NTRIP client per public IP; the recorded CPU and GPU sessions run sequentially. The public service is documented at [Centipede: connecting to the caster](#). The exact source URL, bytes, UTC reception records, connection boundaries and SHA-256 fingerprints accompany each capture. No local coordinate/GGA is sent.

RTCM3 uses preamble D3, a ten-bit payload length and CRC24Q with polynomial $0x1864CFB$. CRC is transport-error detection, not authentication. The implemented receiver extracts GPS LNAV message 1019; station metadata 1005/1006; legacy GPS code 1004; and GPS MSM4/5/6/7 (1074–1077). Other constellation MSM messages are consumed so their terminal flags correctly close the shared observation epoch. Their measurements do not enter the GPS-only solve.

An epoch is keyed by station ID and full GPST time. The multiple-message flag keeps fragments together until a terminal fragment arrives. CRC loss, reconnect, changed epoch before termination and end-of-input produce explicit incomplete records. RTCM has no fragment count or sequence-start flag: a newly connected client cannot prove that it saw the first fragment. This limitation is retained in the event metadata; missing ranges are not reconstructed from station geometry.

The one selected signal per satellite is C1C (L1 C/A), mapped to channel $i = \text{PRN} - 1$. MSM7's finer positive pseudorange takes precedence over legacy 1004 for the same signal. A valid legacy value may fill an unavailable MSM value. All original signal fields are retained; zero, absent or invalid code does not become a solver row. Carrier phase is decoded/preserved where present

but no ambiguity state is estimated by this profile.

3.2 Time, provenance and reproducible execution

Times are unwrapped GPS seconds from 1980-01-06. RTCM week ambiguity is resolved against the recorded reception date, with an explicit GPS-minus-UTC offset of 18 seconds for these sessions. This value is stated by [USNO Earth-orientation values](#); it is a configurable input, not an automatically maintained leap-second service. RINEX epochs already tagged GPST receive no additional UTC offset.

No RTCM 1019 message contains the Klobuchar coefficients. The demonstrated sessions use only the GPSA/GPSB header of the same-day BKG broadcast file `BRDC00WRD_R_20262570000_01D_MN.rnx`:

$$\alpha = (1.2107 \times 10^{-8}, 1.4901 \times 10^{-8}, -5.9605 \times 10^{-8}, -1.1921 \times 10^{-7}), \quad (38)$$

$$\beta = (96256, 81920, -196610, -393220). \quad (39)$$

Its orbital records are *not* loaded during RTCM operation. Every usable satellite orbit/clock arrives in message 1019 from the stream itself. Download URLs, UTC acquisition time and hashes are in the source-data manifests. With no ionosphere header, the model explicitly labels its 5 ns baseline approximation.

Run the included Windows CPU worker from the release directory:

```
python tools/live_satnav.py
--binary bin/cpu/satnav_stream.exe
--url ntrip://caster.centipede.fr:2101/LIENSS
--iono-nav source/live_data/
      BRDC00WRD_R_20262570000_01D_MN.rnx
--seconds 180 --out results/my_live_run
```

The display wraps one command for readability; join it onto one line and keep the navigation-file path continuous. Select the CUDA binary and add `-backend cuda` for the GPU path. Use a new output directory for each run. The dated ionosphere file reproduces this experiment; future-day operation should supply a current header or explicitly select the documented fallback.

The same final received bytes can be replayed with an explicit full-time reference:

```
python tools/live_satnav.py
--binary bin/cpu/satnav_stream.exe
--rtcm source/live_data/
      centipede_lienss_20260914/capture.rtc3
--reference-gpst 1473457283
--iono-nav source/live_data/
      BRDC00WRD_R_20262570000_01D_MN.rnx
--out results/my_recorded_run
```

RINEX input uses `-rinex-obs` and `-nav`; it is explicitly recorded as file replay. The input contract, all optional arguments, live-to-CGK bridge and independent oracle commands are supplied in `docs/LIVE_INPUT.md`, `docs/LIVE_HANDOFF.md` and the source-data README.

3.3 LIVE-R1 time contract and broadcast orbit

A raw observation is $(t_r^{\text{tag}}, P_i, \text{PRN}_i, \sigma_{0i})$. Time is full, unwrapped GPS seconds from 1980-01-06; P_i is the original GPS L1 C/A pseudorange in metres. Receiver time tags and code share one clock convention. GPS-labelled RINEX calendar fields receive no UTC conversion. The final physical reception-time estimate is

$$\hat{t}_r = t_r^{\text{tag}} - \hat{b}_r/c, \quad c = 299\,792\,458 \text{ m s}^{-1}. \quad (40)$$

Here b_r is the estimated receiver clock bias in metres. The raw tag remains in the trace. RTCM week rollover is resolved against an explicit reception-date reference; age tests use full time, never a modulo-week age.

Each GPS ephemeris provides $\sqrt{A}, e, i_0, \Omega_0, \omega, M_0, \Delta n, \dot{i}, \dot{\Omega}$, six harmonic coefficients, t_{oe}, t_{oc} , three clock coefficients, T_{GD} , health and issue identifiers. The selected record has matching PRN, healthy status and IODC mod 256 = IODE. Both clock and orbit age are bounded by 7200 s; a shorter declared half-fit interval further limits orbit age. A nearer unhealthy issue is not replaced by an older healthy issue.

For signal emission time t_s , define

$$A = (\sqrt{A})^2, \quad t_k = t_s - t_{oe}, \quad (41)$$

$$n = \sqrt{\mu/A^3} + \Delta n, \quad M = M_0 + nt_k, \quad \mu = 3.986005 \times 10^{14} \text{ m}^3 \text{ s}^{-2}. \quad (42)$$

Newton's method solves $E - e \sin E = M$:

$$E_{j+1} = E_j - \frac{E_j - e \sin E_j - M}{1 - e \cos E_j}. \quad (43)$$

The implementation reduces M modulo 2π , starts at $E_0 = M$, and requires an increment below 10^{-14} rad within 30 iterations. For $0 \leq e < 1$,

$$\nu = \text{atan2}\left(\sqrt{1-e^2} \sin E, \cos E - e\right), \quad \phi = \nu + \omega, \quad (44)$$

$$u = \phi + C_{us} \sin 2\phi + C_{uc} \cos 2\phi, \quad (45)$$

$$r = A(1 - e \cos E) + C_{rs} \sin 2\phi + C_{rc} \cos 2\phi, \quad (46)$$

$$i = i_0 + \dot{i} t_k + C_{is} \sin 2\phi + C_{ic} \cos 2\phi, \quad (47)$$

$$\Omega = \Omega_0 + (\dot{\Omega} - \omega_e)t_k - \omega_e(t_{oe} \bmod 604800), \quad (48)$$

$$\omega_e = 7.2921151467 \times 10^{-5} \text{ rad s}^{-1}. \quad (49)$$

All angles are radians. Radius harmonics are metres; the other harmonics are angular corrections. These definitions implement the GPS LNAV user algorithm in [IS-GPS-200N, section 20.3.3.4.3](#); they are new GPS-specific additions to this kernel package.

3.4 Satellite clock, emission time and reception axes

Let $x' = r \cos u$ and $y' = r \sin u$. The satellite position in the Earth-fixed axes at emission is

$$\tilde{\mathbf{x}}_s = \begin{pmatrix} x' \cos \Omega - y' \cos i \sin \Omega \\ x' \sin \Omega + y' \cos i \cos \Omega \\ y' \sin i \end{pmatrix}. \quad (50)$$

With $\Delta t_c = t_s - t_{oc}$, the broadcast clock polynomial, eccentric-orbit relativity correction, and L1 code clock are

$$\delta t_{\text{poly}} = a_{f0} + a_{f1} \Delta t_c + a_{f2} \Delta t_c^2, \quad (51)$$

$$\delta t_{\text{rel}} = F e \sqrt{A} \sin E, \quad F = -4.442807633 \times 10^{-10} \text{ s m}^{-1/2}, \quad (52)$$

$$\delta t_{s,L1} = \delta t_{\text{poly}} + \delta t_{\text{rel}} - T_{GD}. \quad (53)$$

The trace stores all three contributions separately. The $-T_{GD}$ sign is specific to this L1 profile. An external differential C1–P1 bias is not invented. Alternate code types are rejected until their signal-bias model is explicit. See [IS-GPS-200N, sections 20.3.3.3.1–20.3.3.3.2](#).

The measured code connects the receiver and satellite clock tags:

$$t_s^{\text{tag}} = t_r^{\text{tag}} - P/c, \quad t_s^{(j+1)} = t_s^{\text{tag}} - \delta t_{s,L1}(t_s^{(j)}). \quad (54)$$

The adapter uses three clock iterations starting at t_s^{tag} . The receiver clock cancels in $t_r^{\text{tag}} - P/c$, because it occurs in both raw quantities. Subtracting the receiver clock again here would double-correct the emission time. This construction follows ESA's [pseudorange-based emission-time algorithm](#).

Earth rotation over the travel time is applied once, before the native solve:

$$R_3(a) = \begin{pmatrix} \cos a & \sin a & 0 \\ -\sin a & \cos a & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (55)$$

$$\mathbf{x}_s^{rx} = R_3(\omega_e \tau) \tilde{\mathbf{x}}_s, \quad \tau = \|\mathbf{x}_s^{rx} - \mathbf{x}_r\|/c. \quad (56)$$

Four range iterations, initially $\tau_0 = P/c$, determine the reception-frame satellite position for the current receiver estimate. The native model never rotates it a second time. The rotation sign and frame convention follow [ESA's reception-frame coordinate construction](#).

The first GSI epoch supplies an independent component check: all eight emission positions match the native RTKLIB trace within 0.000578 m and its clock values within 0.000438 ns. The RTKLIB trace rounds coordinates to millimetres and clocks to 0.001 ns, and excludes T_{GD} from its reported satellite clock. The comparison therefore adds T_{GD} back to Equation (53) for that check only.

3.5 Complete Klobuchar L1 ionosphere

WGS84 ECEF-to-geodetic conversion supplies receiver latitude φ_r , longitude λ_r and ellipsoidal height h . Rotating the line of sight into local ENU gives azimuth A_z clockwise from north and elevation $\mathcal{E}$. A default 10° elevation mask is applied after the receiver geometry has been initialized.

Klobuchar latitude, longitude and elevation arguments are in semicircles: $\varphi_u = \varphi_r/\pi$, $\lambda_u = \lambda_r/\pi$ and $\epsilon = \mathcal{E}/\pi$. Azimuth remains in radians. The complete pierce-point and geomagnetic construction is

$$\psi = \frac{0.0137}{\epsilon + 0.11} - 0.022, \quad (57)$$

$$\varphi_I = \min(0.416, \max(-0.416, \varphi_u + \psi \cos A_z)), \quad (58)$$

$$\lambda_I = \lambda_u + \frac{\psi \sin A_z}{\cos(\pi \varphi_I)}, \quad (59)$$

$$\varphi_m = \varphi_I + 0.064 \cos(\pi(\lambda_I - 1.617)), \quad (60)$$

$$t_{\text{local}} = (43200\lambda_I + t_r^{\text{tag}}) \bmod 86400. \quad (61)$$

With the four broadcast alpha and four beta coefficients,

$$A_I = \max\left(0, \sum_{j=0}^3 \alpha_j \varphi_m^j\right), \quad (62)$$

$$P_I = \max\left(72000, \sum_{j=0}^3 \beta_j \varphi_m^j\right), \quad (63)$$

$$X = \frac{2\pi(t_{\text{local}} - 50400)}{P_I}, \quad (64)$$

$$F_I = 1 + 16(0.53 - \epsilon)^3. \quad (65)$$

Amplitude A_I is seconds and period P_I is seconds. The slant L1 delay, converted to metres, is

$$I = cF_I \begin{cases} 5 \times 10^{-9} + A_I (1 - X^2/2 + X^4/24), & |X| < 1.57, \\ 5 \times 10^{-9}, & \text{otherwise.} \end{cases} \quad (66)$$

Every trigonometric operation on a semicircle value explicitly multiplies by π . The limits on pierce latitude, amplitude and period are part of the published model. See [ESA's Klobuchar equations](#), and GPS LNAV ionosphere parameters in [IS-GPS-200N, section 20.3.3.5.2.5](#).

When broadcast coefficients are unavailable, the adapter explicitly selects $\alpha_j = \beta_j = 0$ and records `klobuchar_zero_coefficients_5ns_baseline`. This still produces $cF_I 5 \times 10^{-9}$ metres of delay; it is not zero ionosphere. For the internet capture, an independently downloaded BKG broadcast navigation header provides the coefficients, with its source hash retained. Receiver ephemerides can still come from the actual RTCM 1019 stream.

3.6 Troposphere, observation variance and native solve

LIVE-R1 uses a specified standard atmosphere with relative humidity $H = 0.7$. For $-100 \leq h \leq 10000$ m and $\mathcal{E} > 0$, define

$$h_0 = \max(h, 0), \quad (67)$$

$$p = 1013.25(1 - 2.2557 \times 10^{-5} h_0)^{5.2568} \quad [\text{hPa}], \quad (68)$$

$$T = 288.16 - 0.0065 h_0 \quad [\text{K}], \quad (69)$$

$$e_w = 6.108 H \exp\left(\frac{17.15T - 4684}{T - 38.45}\right) \quad [\text{hPa}]. \quad (70)$$

The hydrostatic and wet Saastamoinen delays are

$$T_h = \frac{0.0022768p}{(1 - 0.00266 \cos 2\varphi_r - 0.00028 h_0/1000) \sin \mathcal{E}}, \quad (71)$$

$$T_w = \frac{0.002277(1255/T + 0.05)e_w}{\sin \mathcal{E}}, \quad T_d = T_h + T_w. \quad (72)$$

Negative terrestrial heights use sea-level weather and carry a specific status. Outside the height domain, atmosphere is omitted with an explicit diagnostic; no calibrated weather is inferred. These constants and equations match the independent [RTKLIB tropmodel source](#).

The default nominal code uncertainty is $\sigma_{0i} = 3$ m. The executed native profile weights observations by

$$\sigma_i = \frac{\sigma_{0i}}{\max(\sin \mathcal{E}_i, 0.1)}, \quad W_{ii} = \sigma_i^{-2}. \quad (73)$$

CN0 is retained when supplied, without an invented CN0-to-variance calibration. These are nominal independent variances, not measured position errors or protection levels.

The full measurement equation and its single additive correction are

$$P_i = \|\mathbf{x}_r - \mathbf{x}_{s,i}^{rx}\| + b_r - c\delta t_{s,i,L1} + I_i + T_{d,i} + \epsilon_i, \quad (74)$$

$$a_i = c\delta t_{s,i,L1} - I_i - T_{d,i}, \quad P_{i,\text{corr}} = P_i + a_i. \quad (75)$$

The native CPU/CUDA solver estimates $\mathbf{s} = (x_r, y_r, z_r, b_r)^\top$ by weighted least squares. For $\rho_i = \|\mathbf{x}_r - \mathbf{x}_{s,i}^{rx}\|$,

$$r_i = P_{i,\text{corr}} - \rho_i - b_r, \quad H_i = \left((\mathbf{x}_r - \mathbf{x}_{s,i}^{rx})^\top / \rho_i \quad 1 \right). \quad (76)$$

An online Givens QR solve of the whitened system gives $\Delta \mathbf{s}$; $\mathbf{s} \leftarrow \mathbf{s} + \Delta \mathbf{s}$. Rank, iteration, finite-value and residual-fit results remain explicit. Literal ASA/NA selection is unchanged, including whole-word absorption on a selected boundary hit.

3.7 Persistent receiver execution and correction feedback

The persistent native worker receives complete epochs. The preparation layer and native solve form the outer state-feedback map

$$O_n^{(k)} = \text{Prepare}(P_n, t_n^{\text{tag}}, \mathcal{E}_n, s_n^{(k)}), \quad (77)$$

$$s_n^{(k+1)} = \text{NativeSolve}(O_n^{(k)}, s_n^{(k)}), \quad (78)$$

where $\mathcal{E}_n$ denotes the available ephemeris set. Preparation recomputes reception-frame geometry, elevation selection, ionosphere, troposphere and weights from the new position. The default outer limit is eight passes. A valid position requires initialized terrestrial geometry, a converged native solve, and

$$\max(\|x_n^{(k+1)} - x_n^{(k)}\|, |b_n^{(k+1)} - b_n^{(k)}|) < 10^{-3} \text{ m}. \quad (79)$$

At the first receiver epoch, the complete state is zero. Geometry initialization uses all available healthy GPS codes with orbit and clock corrections; the trace explicitly marks atmosphere and elevation as uninitialized. After the first coarse Earth position is solved, the next pass activates the physical models. Reference-station coordinates are never read as seeds. Subsequent epochs start from this worker's preceding valid solution:

$$s_{n+1}^{(0)} = \hat{s}_n. \quad (80)$$

A failed epoch does not overwrite that state. A detected receiver/station change resets the seed. Duplicate or out-of-order epochs do not produce a new fix. This is per-epoch code positioning with a warm start; it does not silently add a motion prior or carrier-phase ambiguities.

The geometric mechanics profile remains a downstream consumer of position results. Its JK state is not injected into satellite orbit/clock corrections or used as a fictitious GNSS observation. The live feedback in Equation (78) closes the formerly missing receiver preparation loop around the existing position kernel.

Separate observable evidence. The journal stores raw measurements, every preparation pass, every native solution and final availability. Correction components retain signs and units. A recorded-file replay, a local timed replay, and a current internet receiver stream are distinct source modes. A remote reference receiver's position is its antenna position; it does not measure the position of the user's computer.

The GSI 120-epoch run starts at ECEF zero, converges on all epochs, and agrees with independent NumPy solves of all 263 converged passes within 1.17×10^{-8} m per state component. The executed weighting differs from RTKLIB's broadcast-variance weighting: their final positions differ by a mean 0.529 m and a maximum 1.589 m. A controlled NumPy re-solve with the same physical corrections and RTKLIB weights reduces those differences to a mean 0.000174 m and a maximum 0.000346 m. This experiment isolates the weighting choice; it is recorded separately from the executed native profile. The full report is `results/real_0759_cpu/independent_comparison.json`.

3.8 Persistent execution and exact records

`satnav_stream` starts once and accepts successive complete requests. Each request contains the epoch ID/time, four-component seed, unchanged ASA/NA/boundary words and up to 32 observation rows. Replies contain status, fit classification, ECEF/clock, formal variance, residuals, iteration count and kernel timing. Both CPU and CUDA dispatch the same SATNAV equations. A

rejected request produces an error record; a Python response timeout closes the worker so a late response cannot be assigned to the next epoch.

Each completed decoded epoch triggers correction preparation and native solves while the socket session is still open. There is no whole-file staging step in the live path. Warm starts carry only the last complete position/clock estimate; they do not introduce velocity filtering or modify raw ranges. Native failure, missing ephemeris and exhausted outer correction iterations publish a null position with an explicit status. Diagnostics still retain the partial iterate. Fit classification remains separate from numerical convergence.

Artifact	Recorded content
<code>capture.rtc3 + sidecars</code>	Raw received body, SHA-256, host times and connection boundaries.
<code>events.jsonl</code>	Decoded original fields, orbital issues, station metadata and incomplete/error records.
<code>trace.jsonl</code>	Raw epoch; every preparation and native pass; final position/clock/time and validity.
<code>prepared/</code>	Exact final-pass CSV geometry/corrections for independent Householder and native batch replay.
<code>summary.json</code>	Source identity, counts, worker binary identity, timing scope and capture outcome.

Only one synchronous callback operates on received bytes. If decoding, disk I/O or positioning becomes slower than delivery, TCP backpressure accumulates upstream. The local demonstration measures a 1 Hz station and is bounded in duration/bytes; it establishes neither continuous service availability nor a hard real-time bound. Socket timeouts, CRC/epoch-loss tests and recorded continuity expose the tested failure semantics. Clock-tag freshness is reported separately from local compute time: this host's UTC clock was about two seconds behind the incoming GPST tags, so a negative tag-to-host interval is not interpreted as negative network latency.

3.9 Current live execution and independent validation

Both final sessions ran on 14 September 2026 UTC (the local Netherlands date crossed into 15 September during the work). Each captured 180 complete one-second epochs using one public caster connection. CPU and CUDA ran sequential sessions: CPU: 21:55:20–21:58:20 UTC; CUDA: 21:58:54–22:01:54 UTC. The source is `caster.centipede.fr:2101/LIENSS`.

Measured quantity	CPU	CUDA
Complete received epochs	180	180
Published native positions	172	175
Initial unavailable epochs (ephemeris warmup)	8	5
Initial ECEF/clock seed	Zero	Zero
Mean 3D separation from advertised ARP (m)	1.644	1.854
Maximum ARP separation (m)	2.232	2.267
Mean 3D difference from RTKLIB (m)	0.649	0.622
Maximum RTKLIB difference (m)	0.969	0.931
Median complete-input-to-position (ms)	2.205	5.947
Maximum complete-input-to-position (ms)	11.611	14.346
All positions emitted during socket capture	Yes	Yes

The native outputs and their timestamps were checked against the exact raw-frame hashes and receive-chunk sidecars. The latency above runs from the host receiving the final required input fragment to publication of a valid position. It includes local decoding and solving; it is not a calibrated radio/network transit time. It excludes unavailable warmup epochs. CUDA's first unavailable live epoch took 2.59 seconds inside processing, including initial device work. A separate persistent-worker regression measured about 1.05 seconds for its first request. CPU is faster for this small 1 Hz workload; these results establish CUDA correctness, not a speedup. The CUDA device was the RTX 5070 Ti Laptop, compute capability 12.0, built with CUDA 12.8 and driver 591.59.

The comparison coordinate is the stream's advertised antenna reference point:

$$(4426043.0455, -89429.1998, 4576296.6447) \text{ m.} \quad (81)$$

It was neither a supplied initial estimate nor an observation. It is not an independently surveyed truth coordinate in this experiment. Consequently the table reports separation from advertised ARP, not certified absolute accuracy. RTKLIB also performs single-point code positioning here; its output quality code is 5, not an RTK fixed-ambiguity solution.

3.10 Measured position series and independent checks

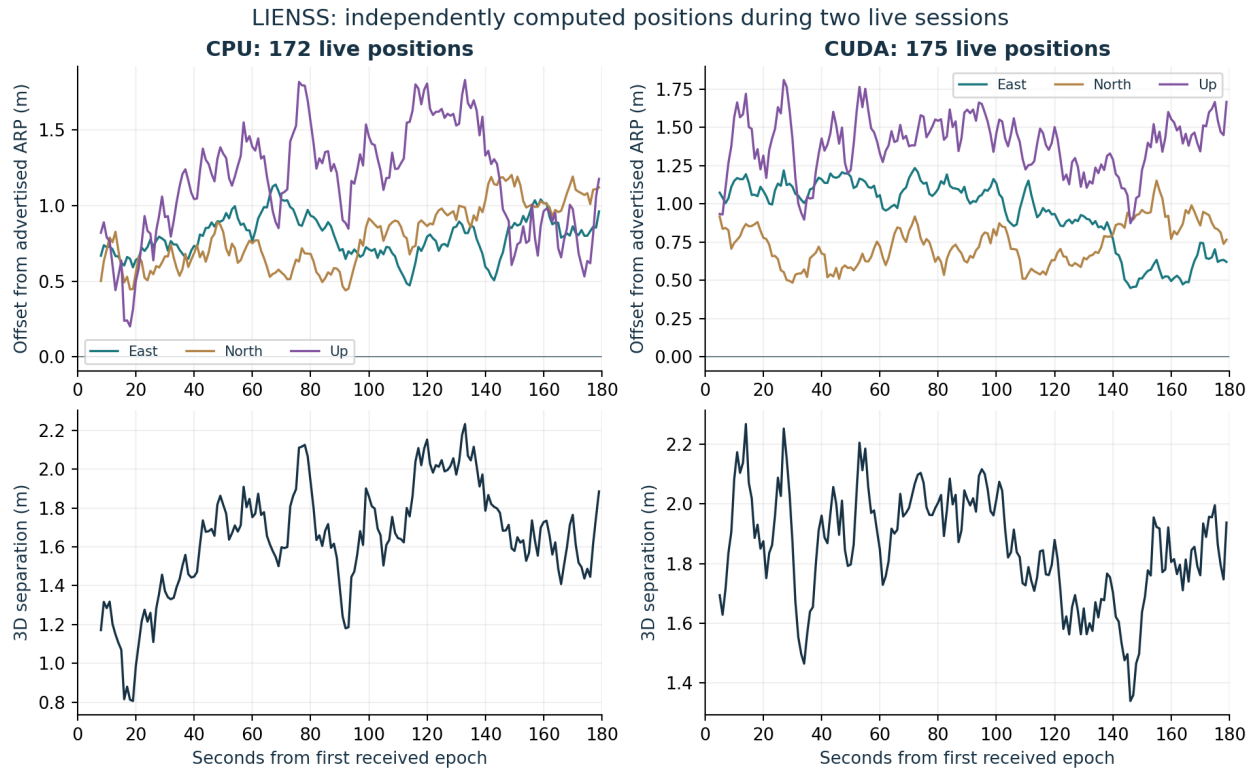


Figure. East/North/Up and three-dimensional separation from the advertised ARP in the two actual live sessions. The blank beginning is ephemeris warmup; no position is fabricated for it. These are different time windows, so their physical differences are not a CPU-versus-GPU numerical comparison.

For that numerical comparison, the same final raw capture was replayed through both backends with the same preparation model and receive order. Both produced 175 positions from 180 epochs; their maximum ECEF/clock component difference was 1.21×10^{-8} m. Live CUDA and its archived-byte replay also passed the same epoch/status/position comparison. `results/live_cross_backend.json` records it.

The independent published RTKLIB `convbin` decoded each capture to RINEX. The final CPU and CUDA decoded satellite/epoch sets matched exactly; 1980 and 2094 common C1C ranges agreed to 0.279 mm, within RINEX decimal rounding. Respectively 280 and 588 ephemeris numeric fields agreed. A separate `pyrtcm` field audit on the five-minute source capture passed 96,880 comparisons. The orbit/clock oracle matched eight satellite positions to 0.578 mm and clocks to 0.000438 ns, limited by its printed output precision. Source hashes and commands are retained.

3.11 Regression, real-model checks and retained state connection

The older GSI recording solved all 120 real epochs from ECEF zero. Its raw physical model and weight-isolation experiment are described with the equations above. Prepared final live CPU observations passed 3,324 independent Householder checks across all 180 epochs, with maximum state-component difference below 8.4×10^{-9} m. All partial/warmup statuses were retained. The persistent CUDA worker processed those same real final-pass inputs under Compute Sanitizer: zero memory errors, 180 responses and 1,048 independent checks.

Current native builds passed all three CTest suites on CPU and CUDA. Python passed 118 test methods. The persistent-interface regression passed 6,861 comparisons per backend, including rejected requests followed by successful recovery. The existing texture/global CUDA paths also passed independent comparison and Compute Sanitizer through `validate_gpu.py -sanitizer`. Analytic/parser tests cover wrong CRC, fragments, missing data, stream truncation, HTTP chunking, reconnection, callback failure, stale epochs, native failure and cold-start rules.

Independent NumPy re-solves verified all 698 converged outer passes across both final live sessions to 1.40×10^{-8} m per component. The physical audit checked 6,284 prepared rows, including original code, correction signs and proof that every chosen 1019 issue had already arrived before its epoch was solved. All 347 published fixes preserve the corrected reception-time equation.

The same live position records also cross the retained CGK-R1 handoff after capture. Only completed positions enter that downstream replay; omitted warmup epochs remain listed in its manifest. ENU is anchored at the first computed position, never at the advertised ARP. Original IDs, ECEF, receiver clock and full time remain attached. Both key layouts are emitted wherever the chart is defined. The stationary demonstration does not require the word state to switch, and its modeled mechanics are not interpreted as a measured receiver velocity. The final CPU bridge retained 172 fixes and listed eight warmup omissions; its native batch position/clock replay exactly matched the stream. CGK replay passed 16,684 field comparisons, and a separate binding audit passed 1,042 checks. The word remained $q = 0$ in this static session. Both keys were emitted for 153 positions; 19 remained inside the chart core. These facts do not assert dynamic switching.

Evidence index. `results/validation_status.json` links current reports and their hashes. `results/network_probes/` contains the independent conversion, raw-field and RTKLIB comparisons. `source/live_data/README.md` records data provenance and third-party attribution. Parent 3.6.1.7 totals in the following chapters are explicitly historical and are not added to these live-session results. The original sibling release remains unchanged.

3.12 Scope of the resulting navigation kernel

This release now contains real raw-data ingestion, satellite ephemeris propagation, explicit physical code corrections, continuous native position/clock computation, and independent real-observation evidence. It is a working single-point GPS L1 receiver backend. The validation uses a static public base station and an older real receiver recording; it does not establish moving-receiver field performance.

The retained CGK-R1 equations provide an explicit geometry/word/mechanics model downstream. Their modeled motion must remain distinguishable from receiver measurements; ASA/NA and JK feedback alone do not add independent observables or cancel unknown GNSS errors. Full ECEF, Up, receiver clock, linear time and both UGTS key layouts remain represented under their existing contracts.

Additional applications can consume the position/time records: maps, logging, spatial queries, digital twins or a later motion estimator. A moving local satnav still needs its own receiver observation source; higher precision needs additional models such as differential corrections or carrier ambiguities. RTK/PPP, Doppler velocity, multi-constellation clock states, map routing and calibrated antenna/code bias estimation are separate implementations. The delivered static-session statistics and formal covariance do not stand in for those missing capabilities.

Use case	Available output or mechanism	Additional work for deployment
Remote position logging	Live ECEF, clock, full time, raw observations and reproducible lineage.	Long-duration operation, service monitoring and application storage.
Local moving satnav	Executable receiver-to-position backend.	Local receiver observations, motion validation and the desired map/routing interface.
Spatial indexing	ENU/Up, chart state, full time and both exact UGTS key layouts.	Application database, spatial extent and query policy.
Digital simulation	Editable literal equations and coupled geometry/mechanics.	Calibrated model parameters and validation against the intended system.
Precise positioning	Raw code and available carrier metadata retained.	Differential/precise corrections, bias models and ambiguity estimation.

4 The completed geometry–state feedback connection

Version 3.6.1.7 defines and implements **CGK-R1**, a coupled continuous and discrete state kernel. Prepared GNSS observations are solved by SATNAV-R1. Their unquantized position estimates become inputs to a persistent modeled mechanical state. Geometry of that state is encoded into the literal ASA/NA and J–K word operations. Their result changes the mechanical operator, and the advanced position supplies geometry for the following step.

The complete executed dependency

Corrected-code observations $\rightarrow$ SATNAV ECEF estimate $\rightarrow$ local target $z_n \rightarrow$ geometry of the carried state $p_{n-1} \rightarrow$ predicate words $\rightarrow$ editable drive/X/J/K equations $\rightarrow q_n \rightarrow$ stiffness and force $\rightarrow (p_n, v_n) \rightarrow$ next geometry.

The actual UGTS handoff publishes the carried J/K state and both the original measurement estimate and the modeled physical state.

The source did not supply complete M2 circle-plus, predicate-encoder or physical eigenmatrix definitions. This release supplies explicit new definitions for all three roles. Their equations, input parameters, units and execution order are given here and in docs/COUPLED_CONTRACT.md. They are new formalizations using the retained primitives, not claims of reconstructing unavailable text.

4.1 State, observations and time identity

For each independently seeded trajectory, retain

$$S = (p, v, q, \phi, t, \rho_{\text{prev}}, \theta_{\text{prev}}, \chi_{\text{prev}}), \quad (82)$$

where $p \in \mathbb{R}^3$ is ENU position in metres, $v \in \mathbb{R}^3$ is velocity in m/s, q is a 32-bit word, ϕ is the separate hoop phase, t is full unwrapped time and χ_{prev} records whether the previous chart exists. Satellite geometry, clock bias, formal covariance and original observation identity remain available in the measurement record outside this state and its compact keys.

Sample n supplies time t_n , target z_n , observation validity $o_n \in \{0, 1\}$, and external force f_n . Define $\Delta t = t_n - t_{n-1} > 0$. Predicates read the old state together with the endpoint observation. The observed target and force are the piecewise constant inputs of the declared discrete observer interval; the resulting mechanical state is labeled t_n . The trace records both interval endpoints, and input rows are never silently reordered.

For a solved ECEF position $\hat{r}_n$, the fixed ENU transform gives

$$z_n = R_{\text{ENU}}(\hat{r}_n - r_0^{\text{ECEF}}). \quad (83)$$

The modeled position p_n is a separate quantity. It is not substituted into the original receiver solution or represented as another independent GNSS measurement.

5 Literal chart motion and an explicit circle-plus

For the old modeled position $p = (E, N, U)$, set $r = \sqrt{E^2 + N^2}$. For $r \geq r_{\text{core}} > 0$, use the retained chart

$$\rho = \log(r/r_0), \quad \theta = \text{atan2}(N, E). \quad (84)$$

The preceding chart belongs to the preceding step's *before* state. Thus the next step compares the newly advanced physical geometry with the geometry that produced that advance. The first step has no chart increment.

For two defined charts, set

$$\Delta\theta = \text{wrap}(\theta - \theta_{\text{prev}}), \quad \Delta\rho = \rho - \rho_{\text{prev}}, \quad \alpha = \arctan(\Delta\theta/\Delta\rho) - a_0. \quad (85)$$

This is literal source OTAN2. A zero pair, zero denominator, missing chart or floating-point range failure retains its explicit status. It is never replaced by quadrant-resolving atan2 while keeping the source name.

5.1 CGK-R1 circle-plus definition

Let $\text{wrap} : \mathbb{R} \rightarrow [-\pi, \pi)$, with the π tie represented by $-\pi$. For $0 \leq w \leq 1$, define

$$a \oplus_w b := \text{wrap}(a + w \text{wrap}(b - a)). \quad (86)$$

This is interpolation along the selected shortest angular arc. It is not Boolean XOR, and associativity is not assumed. At $w = 0$ it selects $\text{wrap}(a)$; at $w = 1$ it selects $\text{wrap}(b)$. Common full-turn changes in either input do not change the selected phase, subject to the declared tie convention.

With $e = z_n - p$, the target bearing $\beta = \text{atan2}(e_N, e_E)$ exists when $o_n = 1$ and the horizontal error is at least r_{core} . The actual blended phase used by the encoder is

$$\psi = \text{wrap}(\beta \oplus_w \alpha + \phi). \quad (87)$$

Both angular inputs must be defined. If they are not, only the alignment predicate is false. Limit, fringe, support, editable Boolean equations and the physical step still evaluate according to their definitions.

5.2 Independent hoop evolution

The hoop has its own supplied angular rate ω_h and evolves as

$$\phi_n = \text{wrap}(\phi_{n-1} + \Delta t \omega_h). \quad (88)$$

Its role stays distinct from geodetic latitude, modeled chart angle and full time winding. It enters the blend explicitly and is preserved in the key tuple.

6 The geometry-to-bit encoders

Each present channel i supplies a unit mechanical direction u_i , angle center β_i , alignment tolerance ϵ_i , projected-error limit L_i , support geometry and fringe width η_i . Parameters and their units are input data; changing them changes the executed model.

For a sphere with center c_i and radius R_i ,

$$s_i(p) = \|p - c_i\| - R_i. \quad (89)$$

A channel may instead select the inherited finite-cone signed-distance function: reduce the point to axial/radial coordinates and take the signed distance to its closed triangular meridian. Slant length, half-angle, apex and unit axis are supplied explicitly. The cone definition is the same one given later in this volume; it is evaluated on the evolving modeled position.

The four Boolean features are exactly

$$A_i^{\text{geom}} = [\psi \text{ defined}] \wedge [|\text{wrap}(\psi - \beta_i)| \leq \epsilon_i], \quad (90)$$

$$L_i^{\text{geom}} = [o_n = 1] \wedge [|u_i^T e| > L_i], \quad (91)$$

$$F_i^{\text{geom}} = [|s_i(p)| \leq \eta_i], \quad (92)$$

$$S_i^{\text{geom}} = [s_i(p) \leq 0]. \quad (93)$$

Channel i occupies bit i ; absent channels contribute zero. The resulting words are named `align`, `limit`, `fringe` and `support`. The separate valid word is all ones for a valid sample and zero otherwise.

6.1 Four actual input equations

The editable JSON supplies drive, input, J and K expressions. The default handoff profile uses

```
"equations": {
  "d": "limit | fringe",
  "x": "(q | d) & support",
  "j": "y & (limit | fringe)",
  "k": "align & ~limit & ~fringe"
}
```

The vocabulary is the exact Boolean algebra $\&$, $|$, $\wedge$, $\sim$, parentheses and word constants $\emptyset/1$. All expressions read the old q . The drive sees six inputs, X sees those plus drive, and J/K additionally see the ASA output y . The compiler enumerates their exact 64-, 128- and 256-bit truth tables. The direct Python reference evaluates the original expression trees rather than these tables.

The source ASA masks and whole-word boundary operation then produce

$$a = x \wedge A \wedge V, \quad n = a \wedge N, \quad h = \text{popcount}(n \wedge B), \quad y = \begin{cases} 0 & h > 0, \\ n & h = 0, \end{cases} \quad (94)$$

followed by the synchronous literal update

$$q_n = [(J \wedge \neg q_{n-1}) \vee (\neg K \wedge q_{n-1})] \wedge V. \quad (95)$$

Absorption determines y ; it does not override the supplied J/K equations.

7 The physical operator supplied by the feedback state

The modeled dynamics have diagonal positive mass M , diagonal damping C , and a supplied symmetric base stiffness K_0 . For each present channel,

$$\kappa_i(q_i) = \kappa_i^{\text{off}} + (\kappa_i^{\text{on}} - \kappa_i^{\text{off}})q_i, \quad (96)$$

$$g_i(q_i) = g_i^{\text{off}} + (g_i^{\text{on}} - g_i^{\text{off}})q_i, \quad (97)$$

$$K(q) = K_0 + \sum_{i \in V} \kappa_i(q_i) u_i u_i^T, \quad (98)$$

$$F(q) = f_n + \sum_{i \in V} g_i(q_i) u_i. \quad (99)$$

Mass entries are kilograms, damping entries kg/s, stiffness entries N/m and force entries newtons. Directions are dimensionless. Signed stiffness and damping are accepted; they can represent an active or unstable model and are reported without silently clipping them to a passive case.

The declared equation of motion over this sample interval is

$$M\ddot{p} + C\dot{p} + o_n K(q_n)(p - z_n) = F(q_n). \quad (100)$$

The newly committed word therefore changes a physical operator, not just an annotation on an independently calculated trajectory. On an unavailable observation, $o_n = 0$ removes the target spring term. The discrete equations, external force, state-dependent force and velocity evolution continue.

7.1 A complete discrete advance

Backward Euler gives the actual linear system solved at every step:

$$G = M + \Delta t C + \Delta t^2 o_n K(q_n), \quad (101)$$

$$b = Mv_{n-1} + \Delta t [F(q_n) + o_n K(q_n)(z_n - p_{n-1})], \quad (102)$$

$$v_n = G^{-1}b, \quad p_n = p_{n-1} + \Delta t v_n. \quad (103)$$

The implementation solves the three-dimensional system with pivoting; it does not form an inverse as its numerical algorithm. The independent reference uses a separate numerical linear solver. Every matrix entry, right-hand side, before state and after state is retained in the trace.

For signed coefficients G can become singular. This is an explicit numerical failure, not a valid state advance. The computed q_n remains recorded; the mechanical state and its valid time stay unchanged. Later rows of that trajectory carry a previous-failure status, while other trajectories continue. Nonfinite diagnostic components are represented as null only on failure records.

8 The physical eigenmatrix and mechanical checks

Because M is positive diagonal and K symmetric, define

$$D(q) = M^{-1/2}K(q)M^{-1/2}. \quad (104)$$

This is the explicit CGK-R1 physical eigenmatrix. It is distinct from the SATNAV observation Jacobian. The native code computes its sorted real eigenvalues and orthonormal eigenvectors, retaining D , K , eigenvectors and a residual. Upper-triangle construction is mirrored to maintain exact floating-point symmetry. The native Jacobi solver scales by the largest matrix entry and uses a pair-relative 2×10^{-15} off-diagonal criterion with at most 64 rotations, preserving small independent modes in matrices with widely separated stiffness scales.

If $Dw_j = \lambda_j w_j$ and $\varphi_j = M^{-1/2}w_j$, then

$$K\varphi_j = \lambda_j M\varphi_j, \quad w_i^T w_j = \delta_{ij}. \quad (105)$$

For $\lambda_j > 0$ the undamped natural angular frequency is $\sqrt{\lambda_j}$ rad/s. For $\lambda_j < 0$, $\sqrt{-\lambda_j}$ is the undamped modal growth rate; a negative value is not reported as a real oscillation frequency. Zero eigenvalues represent neutral stiffness modes. Damping can change the actual transient growth or decay; these are stiffness-mode quantities.

The recorded eigensystem residual is

$$r_\lambda = \max_{a,j} |(DW - W \text{diag } \lambda)_{aj}|. \quad (106)$$

Verification also checks orthonormality and the independently computed spectrum. Signs of eigenvectors and the choice of basis inside repeated eigenspaces are not used as correctness criteria.

8.1 What stability can actually be proved

For a fixed target, zero force, fixed q , $o_n = 1$, $C \geq 0$ and $K \geq 0$, let $E = \frac{1}{2}v^T M v + \frac{1}{2}(p - z)^T K(p - z)$. The implemented backward-Euler step satisfies

$$\begin{aligned} E_n - E_{n-1} = & -\Delta t v_n^T C v_n - \frac{1}{2}(v_n - v_{n-1})^T M(v_n - v_{n-1}) \\ & - \frac{1}{2}(p_n - p_{n-1})^T K(p_n - p_{n-1}) \leq 0. \end{aligned} \quad (107)$$

This follows by substituting the discrete equation and the quadratic difference identity. A changed word changes K and can change stored potential energy; signed coefficients, external forces and moving observations are also outside this fixed-operator inequality. No general convergence claim is substituted for those actual dynamics.

9 Actual SATNAV/UGTS handoff and editable model input

The updated `tools/ugts_handoff.py` invokes the coupled profile by default. It consumes the actual solution and profile files, creates the coupled model input, executes the native backend and independently replays its complete trace. The earlier hold-only behavior is available explicitly as `-legacy-hold`.

Each published record retains the original solver result and a separately labeled modeled state. J/K before, input values and after now come from the executed recurrence. The modeled ENU state is transformed back to ECEF for publication, and its local chart feeds both original 64-bit codecs. Full time, winding, Up, clock bias and original formal covariance remain outside the compact addresses. Failed and unavailable observation rows retain their identity and lineage.

9.1 Inputs are inspectable and editable

`examples/coupled/model.json` declares version 3.6.1.7, profile CGK-R1, the four Boolean equations and a list of trajectories. Each trajectory defines initial time/position/velocity/word/phase, mass, damping, base stiffness, chart and hoop parameters, masks, channel geometries and timed samples. Sphere and cone supports, on/off force and stiffness, directional encoders and the observation sequence are all supplied as data. No geometric-to-bit encoder or physical coefficient is silently inferred from a name.

The native compiler writes four exact CSV schemas: trajectory manifest, channels, samples and equation truth tables. Their order and field meanings are specified in `COUPLED_CONTRACT.md`. The resulting trace has one row per sample, including full geometry, predicate words, ASA tuple, J/K operands, physical matrices, eigensystem and state advance. Hashes bind the supplied model and trace.

9.2 Run the complete connection

From the package directory, using the delivered Windows binaries:

```
bin/cpu/satnav.exe --input examples/demo \
  --out results/my_satnav --backend cpu --verify
python tools/ugts_handoff.py --input examples/demo \
  --run results/my_satnav --backend cpu
```

The coupled tool also runs a directly supplied model:

```
python tools/coupled.py --input examples/coupled/model.json \
  --out results/my_coupled \
  --binary bin/cuda/coupled_kernel.exe --backend cuda
```

Output directories are new for each run. A CUDA thread owns one complete trajectory; it advances that trajectory in time order. Parallelism spans independent trajectories and does not break the carried dependence within a trajectory.

10 Preserved 3.6.1.7 coupled implementation validation

These results were executed for parent version 3.6.1.7 on the local Windows machine.

source/baseline_3_6_1_7_validation.json preserves its index. The unchanged sibling 3.6.1.7 release contains its full run files. These totals are historical evidence, not new 3.6.1.8 live-receiver tests.

Executed check	Result
CPU and CUDA builds / CTest	Passed; three suites per build
Coupled C++ assertions	202 passed
Retained Boolean / SATNAV assertions	5,060,172 / 105,559 passed
Python test methods	70 passed
Retained command-line scenarios	25 Boolean + 11 SATNAV passed
Editable coupled example, each backend	72 steps; 6,984 comparisons
Full coupled suite, each backend	272 trajectories; 3,308 steps
Independent full-suite comparisons	320,876 per backend
Native analytic / causal invariants	20 passed per backend
Deliberate trace corruptions rejected	Eight per backend
Coupled CUDA Compute Sanitizer	Zero errors; full replay passed
Actual SATNAV-to-coupled handoff	256 steps per backend
Independent actual handoff comparisons	24,832 per backend
Separate publication/key/lineage audit	Passed on CPU and CUDA
Retained SATNAV texture/global CUDA	Independent replay and memcheck passed

The comprehensive suite includes 257 trajectories with varying sample counts, so the CUDA run uses three 128-thread blocks with a partial final block. It produces 3,301 advanced states, three deliberately induced singular/overflow failures and four correctly frozen subsequent records. Expected failure records are checked against the independent reference; they are not counted as successful advances.

10.1 Evidence that both directions actually execute

Counterfactual native trajectories share observations while changing the initial word: position and velocity change, and later geometric drive changes. A second pair shares its initial state while changing target geometry: drive and J/K state change, then the physical trajectory changes. Edited Boolean equations also change both state and later predicates. These are executed causal comparisons, beyond agreement between two implementations.

Known free-motion, constant-force and diagonal-spring updates are checked against closed forms of the stated discrete method. Off-diagonal generalized eigenmodes, signed coefficients, missing observations, whole-word absorption, chart-domain cases and numerical failure semantics are exercised. CPU regression checks also retain small independent eigenmodes beside a stiffness of 10^{308} .

11 Coupled handoff evidence and reproducibility

The actual 256-row SATNAV stream produces 253 observation-driven advances and three prediction advances. There are 51 J/K changes across nine distinct word states. At unavailable-observation epochs 250, 251 and 252, modeled motion continues by approximately 2.10713 m, 1.61596 m and 1.24055 m respectively. Residual-alert epoch 253 remains an input observation with its diagnostic intact.

A separate publication audit checks every complete native trace binding, original solver/epoch/residual/clock/variance record, both key encodings, full time, winding, Up and hoop, all 255 state-carry links, all 256 state-dependent stiffness sums and every hash-chain record. Its implementation reads the output independently from the publisher. Original solver records remain distinct from modeled positions.

11.1 Reproduce the full native fixture suite

```
python tools/validate_coupled.py \
  --binary bin/cpu/coupled_kernel.exe \
  --backend cpu --out results/my_cpu_validation
python tools/validate_coupled.py \
  --binary bin/cuda/coupled_kernel.exe \
  --backend cuda --out results/my_cuda_validation
compute-sanitizer --tool memcheck --error-exitcode 99 \
  bin/cuda/coupled_kernel.exe \
  --input results/my_cuda_validation/compiled \
  --out results/my_memcheck --backend cuda
python tools/coupled.py \
  --input results/my_cuda_validation/model.json \
  --out results/my_memcheck_replay --verify results/my_memcheck
```

The replay-only command labels execution as not run because it reads an existing trace; the preceding native command and sanitizer log establish that execution. The native report retains actual backend/device identity and binary/model hashes.

11.2 Toolchain and limits of these results

The builds use MSVC 19.44.35221.0 and NVIDIA CUDA compiler 12.8.61. CUDA executes on an NVIDIA GeForce RTX 5070 Ti Laptop GPU, compute capability 12.0, with driver 591.59. Short build directories avoid this workspace's long Windows scratch paths. Python replay uses NumPy 2.1.2. CPU Address/UndefinedBehavior Sanitizer was not run in this Windows release; its earlier Linux record remains archived.

These checks establish the declared coupled implementation on the supplied finite fixtures. Physical coefficients are editable model inputs; this release has not calibrated them against a measured mechanical system or validated live GNSS data. No general switched-system convergence or measured speedup is inferred. The new blend, encoders and mechanical matrix are fully specified CGK-R1 definitions. They do not purport to recover unavailable original M2 equations.

12 Literal self-reference: the completed recurrence

Version 3.6.1.6 adds the executable profile **SRK-R1**. Its state is a 32-bit word q_t that occurs in its own next-state equation. The old release provided literal ASA/NA and J-K primitives but its observation adapter used a constant hold event. This subversion supplies an explicit feedback composition, editable equation input, CPU/CUDA execution and independent equation replay.

Let $W = \{0, \dots, 2^{32} - 1\}$, and let $V, A, N, B, d \in W$ denote the present, ASA, NA, boundary and external drive words. All complements below are 32-bit complements. The initial state satisfies $q_0 \wedge \neg V = 0$. A trajectory supplies three Boolean expressions E_X, E_J, E_K and a requested number of steps $T > 0$. The complete transition, in evaluation order, is

$$x_t = E_X(q_t, d), \quad (108)$$

$$a_t = x_t \wedge A \wedge V, \quad (109)$$

$$n_t = a_t \wedge N, \quad (110)$$

$$h_t = \text{popcount}(n_t \wedge B), \quad (111)$$

$$y_t = \begin{cases} 0, & h_t > 0, \\ n_t, & h_t = 0, \end{cases} \quad (112)$$

$$J_t = E_J(q_t, y_t, d), \quad K_t = E_K(q_t, y_t, d), \quad (113)$$

$$q_{t+1} = [(J_t \wedge \neg q_t) \vee (\neg K_t \wedge q_t)] \wedge V. \quad (114)$$

The ASA tuple (a_t, n_t, h_t, y_t) is the retained whole-word rule. The bracketed part of (114) is the retained literal J-K equation. The supplied expression bindings and final projection to V are explicit SRK-R1 definitions. For a single active bit they give hold, set, reset and toggle for (J, K) equal to $(0, 0), (1, 0), (0, 1), (1, 1)$ respectively.

Self-reference is an executable dependency

$q_t \longrightarrow x_t \longrightarrow (a_t, n_t, h_t, y_t) \longrightarrow (J_t, K_t) \longrightarrow q_{t+1}$, with q_t also read directly by both J/K expressions and by the final update. Every right-hand side reads the same old state. Only the final line commits the next state.

Boundary absorption sets the whole output word y_t to zero. It does not silently replace the user's J/K expressions with hold: those expressions still evaluate with that zero word. Thus the equations determine the transition even in an absorbed step. Numerical GNSS residual classifications do not gate SRK-R1.

13 Actual equations are input data

The editable input is `examples/self_reference/equations.json`. The root declares version 3.6.1.6, profile SRK-R1, optional default equations, and a `trajectories` array. Each trajectory supplies an ID, initial state, drive, four masks and step count; it may override the equations. For example:

```
{
  "version": "3.6.1.6", "profile": "SRK-R1",
  "equations": {"x": "q | d", "j": "y", "k": "y"},
  "trajectories": [{
    "id": 1, "q0": 0, "drive": 15, "present": 15,
    "asa_mask": 15, "na_mask": 15, "boundary_mask": 0,
    "steps": 6
  }]
}
```

The expression vocabulary is `&`, `|`, `^`, `~`, parentheses, the names `q`, `d`, and, for `J/K`, `y`. The literal `0` means the zero word; `1` means the all-ones word. Decimal state and mask fields remain ordinary unsigned integers. Operator precedence is complement, AND, XOR, OR; parentheses can specify every grouping explicitly.

The expression trees define an exact finite Boolean algebra. They are parsed and evaluated as that algebra; they are not passed to a general-purpose evaluator. Each expression is applied independently to each bit. Whole-word absorption couples the bits through h_t . Arbitrary cross-bit arithmetic, shift operators and continuous-state formulae would require a separately defined extension.

13.1 Exact truth-table compilation

For bits $q_i, y_i, d_i \in \{0, 1\}$, the compiler creates

$$L_X = \sum_{u,v \in \{0,1\}} E_X(u,v) 2^{2u+v}, \quad (115)$$

$$L_J = \sum_{u,v,w \in \{0,1\}} E_J(u,v,w) 2^{4u+2v+w}, \quad (116)$$

$$L_K = \sum_{u,v,w \in \{0,1\}} E_K(u,v,w) 2^{4u+2v+w}. \quad (117)$$

The CPU/CUDA kernel reads the corresponding table bit. For the example above, $(L_X, L_J, L_K) = (14, 204, 204)$. For the latch with $K = 0$, the tuple is $(14, 204, 0)$. Every allowed expression has a truth table on these inputs, and every Boolean function on these inputs is expressible in this vocabulary. The compilation is exact: each possible input assignment is evaluated, and the native table read retrieves that assignment's value.

14 Fixed points, cycles and complete finite execution

During one trajectory, equations, drive and masks are constant. They define a deterministic function $F : Q_V \rightarrow Q_V$, where

$$Q_V = \{q \in W : q \wedge \neg V = 0\}, \quad |Q_V| = 2^{\text{popcount}(V)}, \quad q_{t+1} = F(q_t). \quad (118)$$

Proposition 14.1 (Closure and eventual periodicity). *Every computed state belongs to Q_V . Every infinite orbit of this fixed-input transition is eventually periodic. A fixed point is a cycle of length one.*

Proof. The input requires $q_0 \in Q_V$ and the final AND with V places every successor in Q_V . Among $|Q_V| + 1$ consecutive states at least two are equal. If $q_i = q_j$ for $i < j$, determinism gives $q_{i+k} = q_{j+k}$ for every $k \geq 0$. This proves periodicity after the first repeated state; it does not assert that the period is one. $\square$

The implementation emits every requested step, including those after a repeat. Independent replay records a fixed point, a cycle, or no repeat observed within the requested budget. The last case is finite evidence only; it does not imply divergence or the absence of a later repeat. With 32 active bits the general state bound is 2^{32} , so this theorem alone gives no short runtime bound.

14.1 Worked exact trajectories

Set $V = A = N = d = 15$, $B = 0$, $q_0 = 0$, and $E_X = q \vee d$, $E_J = y$. For the latch $E_K = 0$, $x = a = n = y = 15$ and

$$q_{t+1} = y \vee q_t = 15, \quad (q_0, q_1, q_2, \dots) = (0, 15, 15, \dots). \quad (119)$$

For toggle $E_K = y$, $J = K = 15$ and

$$q_{t+1} = q_t \oplus 15, \quad (q_0, q_1, q_2, q_3, \dots) = (0, 15, 0, 15, \dots). \quad (120)$$

With $B = 8$ and $E_J = E_K = y$, $h = 1$ absorbs all of y and the state holds. With the same absorbed word but $E_J = d$, $E_K = 0$, the state becomes 15. These outcomes follow directly from the supplied equations and expose the difference between output absorption and state reset.

Synchronous evaluation matters. For one active bit with $E_J = \neg q$ and $E_K = q$, the result is toggle. Both expressions use q_t , never a partially updated state. State projection preserves inactive bits as zero, including when an input expression uses the all-ones constant.

15 Native execution and independent equation replay

The shared `include/self_reference.hpp` implements the truth-table and literal transition operations. The CPU walks each trajectory in step order. CUDA assigns one thread to each independently seeded trajectory, then executes its recurrence sequentially within that thread. No ordering between independent trajectories is needed; different trajectories never share a mutable state.

The compilation interface is a decimal CSV with columns

```
trajectory_id,q0,drive,present,asa_mask,na_mask,boundary_mask,
x_lut,j_lut,k_lut,steps
```

The header is physically one line. Every trace row retains

```
trajectory_id,step,before,drive,x,asa,na,hits,output,j,k,after
```

The native run also records backend, device, timing and trace lengths. The Python reference evaluates the original expression trees on whole integer words; it does not use the compiled truth tables as its transition oracle. It compares every trace field, row count and trajectory ordering, and records equation/input provenance. This cross-check can detect table compilation, transition and output corruption; it is finite execution evidence rather than a machine-checked proof.

15.1 Reproduce the equation-input run

From the package directory, build CPU and run:

```
cmake -S . -B build_cpu
cmake --build build_cpu --config Release --parallel 2
python tools/self_reference.py \
  --input examples/self_reference/equations.json \
  --out results/selfref_local \
  --binary build_cpu/Release/self_reference.exe --backend cpu
```

For single-configuration systems the binary is `build_cpu/self_reference`. For CUDA, enable `SATNAV_ENABLE_CUDA=ON`, choose the GPU binary and use `-backend cuda`. Each run uses a new output directory. The JSON expressions are editable without recompiling C++ or CUDA; the compiler regenerates their tables.

15.2 Relation to the existing SATNAV and UGTS equations

SATNAV-R1 retains its corrected-code equations, QR solve, OTAN2 source ratio, both 64-bit codecs and full coordinate/time records. The earlier observation adapter's hold event is retained by the explicit legacy option. SRK-R1 is the retained formal word state-feedback computation and runs without a GNSS residual admission condition. The new bindings are identified as definitions of this release. CGK-R1 now provides explicit new circle-plus, encoder and physical-matrix definitions and connects the observation path, as detailed in Section 4. The unavailable earlier M2 definitions are not claimed to have been reconstructed.

16 SRK-R1 baseline validation, version 3.6.1.6

All entries here concern the executable 3.6.1.6 source on the current Windows machine. The 3.6.1.5 preparation record is preserved separately under `results_3_6_1_5/`. Finite comparisons check the requested trajectories; the closure and eventual-periodicity argument is given separately above.

Executed check	Result
CPU CMake build and CTest	Passed, 2 test programs
Retained SATNAV C++ checks	105,559 passed
New literal feedback C++ checks	5,060,172 passed
Python test methods	31 passed
Native command-line scenarios	25 feedback + 11 SATNAV passed
Editable six-trajectory example	34 transitions, 408 exact field comparisons per backend
257-trajectory stress input	4,225 transitions, 50,700 exact field comparisons per backend
CPU and CUDA equation replay	Both passed sample and stress inputs
Feedback GPU Compute Sanitizer	Zero memory errors; sanitized trace replay passed
Retained SATNAV independent reference	256 epochs, 4,828 comparisons passed
SATNAV CUDA texture/global paths	Both passed independent comparisons and memcheck
CPU sanitizers in this Windows run	Not run; earlier Linux evidence archived

16.1 Recorded toolchain and device

MSVC 19.44.35221.0, Visual Studio 2022 BuildTools; NVIDIA CUDA compiler 12.8.61; NVIDIA GeForce RTX 5070 Ti Laptop GPU, compute capability 12.0, driver 591.59. The source targets `sm_120` and `compute_120`. Device, runtime, allocation sizes and kernel-only times are retained in each native run record. The short external build directories used on this machine avoid Windows compiler scratch-path limits caused by the long workspace path.

The stress input uses seed 3616, varying 1–32 steps per trajectory, full 32-bit operands and 257 trajectories so that the 128-thread CUDA blocks include a partial final block. Direct AST replay checks all 12 trace columns. The Python suite also rejects deliberate mutations of every trace column, missing/extra rows and reordered rows. The C++ checks include independent bit oracles and exhaustive small domains.

16.2 Evidence and delivered implementation

`source/baseline_3_6_1_6_validation.json` indexes the archived baseline evidence. The exact equation JSON, compiled CSV, complete native/reference traces and their hashes are retained. The baseline delivered Windows CPU and CUDA executables under `bin/windows/`; CUDA execution uses the installed NVIDIA runtime and driver. No measured speedup is inferred from these small fixtures. The PDF is built with Tectonic 0.17.0 and visually checked in that release. Source and delivered artifacts were listed in the regenerated SHA-256 manifest.

17 The imported UGTS and aTOMos contracts

The supplied prototype defines a finite, query-first state-and-relation package. It includes cone/-sphere relations, a bounded translational sweep, log-polar metric and kinematics, explicit time and winding, separate topology profiles, finite grammar, 64-bit key codecs and audited uncertainty/memory claims. Its authoritative handoff is

support → compatibility → guard → verified event → transition → lineage. (121)

This order is retained in the downstream satnav event adapter. [U, pp. 1, 9–12]

17.1 What U does not define

U contains no satellite ephemeris propagation, receiver clock observation equation, code-delay correction model, geodetic datum or GNSS position estimator. These are new SATNAV-R1 definitions, grounded in primary GNSS references rather than attributed to the earlier prototype. [N01]

The base aTOMos contracts retained here are the four-field ASA word rule, explicit JK bit semantics, source-ratio OTAN2, typed unavailable values, immutable input snapshots and ordered result/provenance records. The new matrix in the receiver solve is the explicit *GNSS design matrix*. It is not a silently completed physical beamforming/eigenmatrix from an earlier profile. [A, retained M2 project contract]

17.2 Source figures and key table

U's Figure 4 on printed p. 5 keeps the half-turn state bundle distinct from the orientation-reversing Klein quotient. Its Table 1 and Figure 6 on printed p. 8 keep contiguous packing distinct from MSB-round-robin Morton interleaving. Figure 8 on printed p. 10 returns the result to the canonical event sequence. All three distinctions survive the specialization.

The complete earlier 50-operator catalog remains a source catalog, not a claim that every operator is necessary for positioning. The delivered helpers implement the subset selected below. A source mechanism omitted from the default satnav path is identified rather than repurposed without a declared meaning.

18 Receiver state and source-symbol separation

The authoritative positioning state is

$$z_e = (x_e, y_e, z_e, b_e) \in \mathbb{R}^4, \quad b_e = c \delta t_e, \quad (122)$$

where the first three components are ECEF metres and the fourth is receiver clock bias in metres. Clock seconds are b_e/c with $c = 299792458$ m/s in this profile. This is the standard code-positioning unknown set. [G1]

Quantity	Role	Representation
r	Receiver location	Three unquantized ECEF doubles.
b	Receiver clock bias	Metres, not a branch bit.
P_i	Code observation	Metres, after the explicitly recorded correction convention.
ρ	Local log-radius	UGTS chart quantity, not the code range.
θ	Local angular coordinate	$\text{atan2}(N, E)$ in a fixed ENU chart.
ϕ_h	UGTS hoop phase	Separate supplied periodic field; not silently geographic latitude.
$\varphi_g, \lambda_g, h_g$	Geodetic coordinates	Latitude, longitude and ellipsoidal height.
X, w	Tick and winding	Full integer time identity retained beside its modular key field.
K_C, K_M	Quantized addresses	Two distinct 64-bit codecs, not complete navigation states.

The source's lower-case hoop ϕ is retained as ϕ_h to avoid colliding with geodetic latitude. The earlier aTOMos phase alias is resolved through a named adapter: OTAN2 receives increments of local ρ, θ , while the key retains its separate UGTS hoop field. No 3D coordinate is silently removed. [U, p. 2][N02]

The authoritative records also include epoch identity, time scale, satellite/channel mapping, input/product provenance, statuses and uncertainty assumptions. All fit results preserve the full ECEF estimate and clock value outside the compact key.

19 The corrected-code observation model

A decoded code measurement can be modelled as a geometric range, receiver/satellite clock terms and selected propagation/instrumental delays. SATNAV-R1 accepts an upstream additive correction a_i and defines

$$\tilde{P}_i = P_i + a_i = \|s_i - r\| + b + \varepsilon_i. \quad (123)$$

This is the reduced standard code-positioning model after the declared upstream terms are accounted for. [G1, G4]

For the convention

$$P_i = R_i + b - c\delta t_i^{\text{sat}} + T_i + I_i + B_i + \varepsilon_i, \quad (124)$$

the matching additive correction is

$$a_i = c\delta t_i^{\text{sat}} - T_i - I_i - B_i. \quad (125)$$

The adapter must define the exact code, satellite clock convention, relativistic clock terms and hardware/group biases it uses. The kernel adds the supplied term *once*. It does not repeat an already applied correction or invent a missing ionosphere/troposphere model.

19.1 Satellite coordinates and Earth rotation

s_i is the satellite position at signal emission expressed in the Earth-fixed axes associated with reception. The upstream stage must perform the necessary transmit-time and frame handling consistently. A satellite coordinate computed in emission-time ECEF axes cannot simply be treated as already in the reception axes. [G2]

The solver keeps these supplied positions and corrections fixed during its four-state iteration. The profile is therefore an adapter boundary for prepared code data, not a complete broadcast-navigation decoder. Raw receiver or RINEX values require an upstream modelling adapter before using the named CSV contract. [N01]

19.2 One clock system

There is exactly one receiver-clock unknown. Unaligned measurements from multiple GNSS time systems generally need additional bias treatment; this release does not estimate a hidden inter-system offset. Use compatible code/clock data, or supply and document an upstream alignment under a separate profile.

20 Input epochs, channels and aTOMos selection

Each independent epoch binds its receive-time label, approximate initial state and 32-bit channel masks. Each observation row points to that epoch and names a unique channel in $0, \dots, 31$. Channel IDs, not row order, define mask bits.

Let V contain present channels and X contain channels marked ready. The retained word selection is

$$a = X \wedge M_{\text{ASA}} \wedge V, \quad (126)$$

$$n = a \wedge M_{\text{NA}}, \quad (127)$$

$$h = \text{popcount}(n \wedge M_{\partial}), \quad (128)$$

$$y = \begin{cases} 0, & h > 0, \\ n, & h = 0. \end{cases} \quad (129)$$

The solver uses precisely the channels selected by y . These four fields are emitted unchanged in the result. The default ASA/NA masks are all ones and the default boundary is zero. [A, retained M2 project contract] [N04]

20.1 Whole-word behavior remains literal

A boundary bit intersecting the selected word absorbs the entire epoch group. It does not mean “drop only this satellite.” To explicitly omit an individual observation, use a selector bit, a not-ready flag or an upstream policy with its own provenance. Replacing the boundary behavior with per-bit clearing would change the base engine.

A remaining support of fewer than four channels returns T00_FEW. Four channels must still provide a numerically independent design matrix. A full bit count cannot prove observability by itself. The new GNSS solver checks the resulting matrix rather than treating POPCNT as a complete position solution. [G1][N03]

20.2 Timeline and schema consistency

An epoch identifier binds rows from one receiver event. The time column is full unwrapped GPST seconds under the supplied upstream convention; modular UGTS time is added later. Duplicate epoch IDs, duplicate channels or duplicate satellite IDs within an epoch are errors. Correcting a false upstream time label is not something a CSV parser can infer from equal strings.

The current CSV reader uses finite bounded numeric fields and explicit nonnegative IDs. Physical correction provenance belongs to the input profile; metadata consistency is not a physical synchronization measurement.

21 The weighted positioning step

For current estimate $z^{(k)} = (r^{(k)}, b^{(k)})$, define

$$R_i^{(k)} = \|r^{(k)} - s_i\|, \quad (130)$$

$$v_i^{(k)} = \tilde{P}_i - R_i^{(k)} - b^{(k)}, \quad (131)$$

$$H_i^{(k)} = \left[\frac{x^{(k)} - s_{ix}}{R_i^{(k)}}, \frac{y^{(k)} - s_{iy}}{R_i^{(k)}}, \frac{z^{(k)} - s_{iz}}{R_i^{(k)}}, 1 \right]. \quad (132)$$

Linearization gives $H\Delta z \simeq v$. With declared independent code standard deviations $\sigma_i > 0$, whiten the rows as

$$A_i = H_i/\sigma_i, \quad d_i = v_i/\sigma_i, \quad \Delta\hat{z} = \arg \min_{\Delta} \|A\Delta - d\|_2^2. \quad (133)$$

Then update $z^{(k+1)} = z^{(k)} + \Delta\hat{z}$. The nonlinear model and iterative linearization follow the code-positioning references; the QR implementation is the explicit numerical choice of this specialization. [G1, G4][N03]

21.1 No normal-equation update

The C++/CUDA implementation uses streaming Givens QR on the whitened rows. It does not form and invert $A^T A$ for the update. This avoids squaring the design matrix's condition number as part of the solution method. The independent Python reference uses a different Householder QR implementation on the full row matrix.

All four unknowns use metres, including the clock bias. A floating-point conversion to seconds is applied only to a displayed/consumer clock offset. This keeps a factor of the speed of light out of the fourth Jacobian column.

The default iteration stops when both position-increment norm and absolute clock increment are at most 10^{-4} m, with at most 12 steps. These are numerical convergence controls, not a statement that real measurements yield 0.1 mm accuracy. The convergence domain depends on the input model and initial approximation.

22 Streaming QR, rank and formal covariance

The kernel maintains an upper-triangular 4×4 matrix R and a transformed right-hand side z . Each whitened row is appended by rotations that eliminate its components against the current triangular diagonal.

For diagonal R_{jj} and current row component a_j , let

$$\gamma = \text{hypot}(R_{jj}, a_j), \quad c_j = R_{jj}/\gamma, \quad s_j = a_j/\gamma. \quad (134)$$

The same rotation is applied to the remaining row and right-hand-side entries. A zero pair needs no rotation. The result is solved by back substitution.

22.1 The implemented rank screen

The retained check is

$$|R_{jj}| > \tau \max_k |R_{kk}| \quad \text{for all } j, \quad \tau = 10^{-10}. \quad (135)$$

This is a declared unpivoted triangular-diagonal screen. It detects the tested rank-deficient cases, but is not an SVD condition-number certificate or a proof of well-conditioned geometry for all possible inputs. A failed screen produces `RANK_DEFICIENT`, not a coordinate filled with zeros. [N03]

22.2 Covariance is tied to the assumed sigmas

After convergence the Jacobian is rebuilt at the final estimate. From R^{-1} , compute the diagonal of

$$P = (H^T W H)^{-1} = R^{-1} R^{-T}, \quad W = \text{diag}(\sigma_i^{-2}). \quad (136)$$

The implementation obtains this from triangular solves, not a dense matrix inverse of $H^T W H$. It returns coordinate/clock variances in squared metres. This is formal uncertainty under the declared local linear and noise model. [G5]

It is not measured error against truth, and not a guaranteed bound on faulty or mis-modelled measurements. The independent fixture comparisons check both estimates and these formal diagonal values; true-error metrics are separately computed only because the artificial fixture has a known generating position.

23 Solver status and residual diagnostics

Convergence, residual fit and the later UGTS event are separate fields.

Solver status	Meaning
CONVERGED	Numerical increment thresholds reached with valid final calculations.
MASK_ABSORBED	Selected word intersected the retained whole-word boundary.
TOO_FEW	Fewer than four selected observations.
RANK_DEFICIENT	Triangular rank/covariance check failed.
ITERATION_LIMIT	The selected number of iterations did not satisfy convergence.
NUMERIC_FAILURE	Invalid range, nonfinite result or unusable numeric intermediate.

For a converged solution, preserve every selected post-fit residual $r_i = \tilde{P}_i - \|s_i - \hat{r}\| - \hat{b}$, and report

$$\text{RMS} = \sqrt{\frac{1}{m} \sum_i r_i^2}, \quad S = \sum_i (r_i/\sigma_i)^2, \quad Z_{\max} = \max_i |r_i/\sigma_i|. \quad (137)$$

The default comparison $Z_{\max} \leq 6$ is a named residual budget. It is not a leverage-normalized test or a claimed false-alarm probability. The solver does not automatically discard a channel and retry until this flag passes. [N09]

Exactly four observations give NO_REDUNDANCY, even if their residuals are small. For more observations, the flag is WITHIN_RESIDUAL_BUDGET or RESIDUAL_ALERT. A single gross-outlier fixture converges numerically but produces the latter alert; the data record retains its large actual position error.

RAIM/ARAIM, HPL/VPL, continuity risk, certified spoof detection and navigation-action authority are not supplied by these flags. A valid solution record can be useful for replay and analysis without representing those additional guarantees.

24 Local ENU and the UGTS log chart

The receiver solution remains three-dimensional. For a declared geodetic anchor $(\varphi_0, \lambda_0, h_0)$, convert ECEF differences to East/North/Up using

$$R_{ENU} = \begin{bmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{bmatrix}. \quad (138)$$

Then $(E, N, U)^T = R_{ENU}(\mathbf{r} - \mathbf{r}_0)$. The transform is an explicit GNSS/geodesy addition, separate from the source's abstract local chart. [G3]

With horizontal radius $r_h = \sqrt{E^2 + N^2}$, use

$$\rho = \ln(r_h/r_0), \quad \theta = \text{atan2}(N, E). \quad (139)$$

Up remains a separate value. A declared core radius returns `origin_core` instead of evaluating a singular logarithm. Out-of-range ρ is not automatically wrapped into a different geographic position. [U, pp. 3–4][N02]

24.1 Metric and finite steps carried from U

For the planar chart,

$$ds^2 = r_h^2(d\rho^2 + d\theta^2), \quad \Delta r_h = r_h(e^{\Delta\rho} - 1), \quad (140)$$

$$J = r_h \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (141)$$

These identities explain the local representation; the new satellite-range solve still uses full ECEF coordinates. Quantizing this chart does not improve the original measurement precision or create a third coordinate. [U, p. 4]

The geodetic utility uses the declared WGS84 ellipsoid constants $a = 6378137$ m and $1/f = 298.257223563$. Its height is ellipsoidal height, not an unprovided geoid-corrected height above mean sea level. The reference tests its conversions and inverse relation over a finite set of points. [G6]

25 OTAN2 in satnav: preserve the observable

The inherited literal operator remains

$$\text{otan2}_s(\Delta\theta, \Delta\rho) = \arctan(\Delta\theta/\Delta\rho), \quad \Delta\rho \neq 0, \quad (142)$$

with a separately supplied reference-axis subtraction when selected. This is a *log-chart motion slope*. It is not used to solve pseudoranges, and is not relabeled as the receiver's geographic heading or satellite azimuth. [A, retained M2 project contract]

The passive adapter compares successive defined local chart records in the same frame. It uses an explicitly named principal-difference policy for $\Delta\theta$; that policy does not reconstruct missed revolutions. Full linear time and the chosen phase convention are stored with the observation.

25.1 Missing direction remains a status

A first observation has no previous chart increment. A zero pair has no direction. A pure angular increment has an undefined literal ratio. The adapter records `first_observation`, `zero_increment` or `ratio_undefined` rather than inventing a perfect zero angle. A sequence with non-increasing time is identified instead of being silently reordered into a new physical history.

A change in the axis alone changes the raw residual by the opposite angle. A common positive scaling of both increments preserves the ratio, and simultaneous reversal also preserves the literal ratio while reversing the actual directed motion. These are inherited operator laws, not new navigation-sensor guarantees.

25.2 Literal J/K and the explicit feedback profile

The retained equation is

$$q^+ = (J \wedge \neg q) \vee (\neg K \wedge q). \quad (143)$$

SRK-R1 binds J and K to supplied expressions that read the old state and literal ASA output, then carries q^+ into the next step. The full recurrence, expression input and execution semantics are given in Section 12.

The default observation adapter now invokes CGK-R1 and publishes its actual carried state. The earlier hold event remains available through the explicit legacy option. CGK-R1 supplies new circle-plus, encoder and mechanical-operator definitions in Section 4; the unavailable original M2 forms are not claimed to have been recovered.

26 Paired supports and the canonical UGTS event

This section records the retained observation-only adapter, selected with `-legacy-hold`. The default coupled handoff is specified in Section 4; it retains residual diagnostics without using this earlier admission rule to override its equations.

U normalizes the two spherical supports as

$$f_L(x) = \|x - c_L\| - R_L, \quad f_R(x) = \|x - c_R\| - R_R, \quad (144)$$

with a profile selecting union, intersection or either-with-tag. The overlap flag is retained separately. [U, p.3]

SATNAV-R1 uses the union of two *declared local processing domains* for its worked event adapter. With supplied deterministic position-radius allowance u_p , its per-sphere margin is

$$m_j = R_j - \|x - c_j\| - u_p. \quad (145)$$

Admission means at least one margin is nonnegative. This is a geometric/query policy for the supplied local frame, not a claim that the true receiver error is bounded by u_p . Its origin is explicitly recorded. [N09]

The corresponding handoff is implemented in order:

UGTS stage	Satnav realization
Support	Candidate lies within at least one declared local support under the supplied allowance.
Compatibility	Fixed frame and increasing unwrapped time; supported estimator-clock convention.
Guard	Residual interval compared with a named metre-domain budget.
Verified event	Candidate satisfies those computational/model conditions with residual redundancy.
Transition	Publish an observation record only. No external control command.
Lineage	Source/profile IDs, full time, source keys and the preceding record digest.

For the demo, $u_p = 5$ m, the two support radii are 10,000 m, and centres are (0, 0, 0) and (100, 100, 0) ENU metres. These are labelled worked parameters, not values extracted from the attached prototype or a real site survey.

The phrase `POSITION_CANDIDATE_VERIFIED` means verified under this declared computational handoff. It does not supply certified GNSS integrity or prove the physical correctness of the upstream observations.

27 Finite cones and translational sweeps retained

For the source's slant length $T > 0$ and half-angle $0 < \alpha_c < \pi/2$,

$$h = T \cos \alpha_c, \quad R = T \sin \alpha_c. \quad (146)$$

With apex c , unit axis a and query x , set

$$z = a \cdot (x - c), \quad q = \|(x - c) - za\|. \quad (147)$$

The exact primitive relation is the signed meridian distance to the triangle $(-R, h), (R, h), (0, 0)$, with negative interior and zero boundary. The implementation computes distances to the three boundary segments and uses the source's inside test $0 \leq z \leq h, q \leq z \tan \alpha_c$. [U, pp. 2–3]

These are optional UGTS query helpers in the archive. A finite cone is not silently substituted for the satellite pseudorange sphere or a physical antenna response. The default receiver solve uses the actual distance-to-satellite equations already stated.

For a fixed-orientation cone translated along a segment of length L , $n \geq 2$ uniform samples and sample minimum m_n , retain

$$m_n - \frac{L}{2(n-1)} \leq \inf_{u \in [0,1]} d_C(x - s(u)) \leq m_n. \quad (148)$$

The source derives this using the one-Lipschitz distance response to translation and the nearest sample spacing. It is not a general exact SDF for arbitrary rotated sweeps. [U, p. 3]

27.1 Numerical and nesting distinction

The helper evaluates this analytic interval in ordinary floating arithmetic. It does not implement directed rounding or a proven machine-error enclosure. The interval's *width* decreases with sample spacing. Intervals from arbitrary nonnested sample grids need not be nested, so the implementation does not claim that stronger property. [N07]

The source's constraint-release and hinge-torque mechanisms remain separately typed geometric/mechanical profiles. Removing a constraint is not an automatic improvement to satellite observability, nor does the existence of a hinge model supply a physical receiver trajectory. They are not inserted into the default position estimator.

28 Jitter intervals and the explicit source correction

U defines $f_j = f + \epsilon_j \sigma$ with $\sigma \in \{-1, +1\}$, then prints the equality

$$[f_j - \epsilon_j, f_j + \epsilon_j] = [f - \epsilon_j, f + \epsilon_j]. \quad (149)$$

That equality is generally false for nonzero ϵ_j . This edition preserves its source location and explicitly separates the two interpretations rather than silently changing the supplied page. [U, p. 5][N06]

28.1 Two valid, different interval statements

The perturbation family obeys

$$f_j \in [f - \epsilon_j, f + \epsilon_j]. \quad (150)$$

If f_j is the available perturbed observation and $|f_j - f| \leq \epsilon_j$, then

$$f \in [f_j - \epsilon_j, f_j + \epsilon_j]. \quad (151)$$

Both are valid enclosure statements, but the interval centres differ. The helper returns both named intervals and a test verifies that they are not falsely equated. The source's example centred around f is treated as the perturbation-family interval.

No jitter is added to the GNSS code observations or to solved ECEF coordinates. The deterministic bit can be metadata or a separate query perturbation, not a method for manufacturing measurement precision or removing multipath.

28.2 Metre-domain guard in the legacy event adapter

For the largest absolute post-fit residual $m = \max_i |r_i|$ and a *supplied* absolute residual perturbation allowance u_r , form

$$[\max(0, m - u_r), m + u_r]. \quad (152)$$

A budget B is met when the upper endpoint is at most B , exceeded when the lower endpoint is greater than B , and otherwise unresolved. This is a conditional model classification. The demo uses $u_r = 0.1$ m and $B = 20$ m as stated fixture parameters. It does not identify nominal σ_i with a hard worst-case bound.

29 Two exact 64-bit key layouts

U's field widths and layouts are retained exactly:

Field	Bits	Contiguous range	Domain
Log-radius index	20	63:44	Declared $\rho \in [-20, 0]$
Chart angle	18	43:26	Periodic $[0, 2\pi)$
Modular tick	14	25:12	$0, \dots, 16383$
Hoop phase	12	11:0	Periodic $[0, 2\pi)$

The contiguous word is

$$K_C = (q_\rho \ll 44) \vee (q_\theta \ll 26) \vee (q_X \ll 12) \vee q_{\phi_h}. \quad (153)$$

The Morton word emits source bits in MSB round-robin order

$$\rho_{19}, \theta_{17}, X_{13}, \phi_{11}, \rho_{18}, \theta_{16}, X_{12}, \phi_{10}, \dots \quad (154)$$

until each source field is exhausted. They are intentionally different encodings. [U, pp. 7–8]

29.1 The source's exact golden pair

For (949111, 0, 1920, 227), the new tests recover

```
contiguous: e7b77000007800e3
Morton:    88823bb88099128b
```

Both decode to the same quantized tuple. Random round trips and prefix-refinement bounds are tested separately from the new positioning arithmetic.

29.2 Address identity is not state identity

The 64-bit key has exactly 2^{64} possible words. A key does not encode arbitrary clock bias, full linear time, Up, formal covariance, correction products or lineage. Those remain separate full fields in the event record. The 12-bit field is the named hoop phase, not a simultaneously hidden arbitrary matrix and grammar payload. [U, pp. 8–9][N05]

The address construction follows the position estimate; it is not used to quantize satellite coordinates or code ranges before the receiver solve. A matching key is an index match, not proof that two entire navigation events are identical.

30 Quantization, error and explicit origin handling

SATNAV-R1 declares nearest-node radial quantization over $[-20, 0]$:

$$h_\rho = \frac{20}{2^{20} - 1}, \quad q_\rho = \text{round}_+ \left(\frac{\rho + 20}{20} (2^{20} - 1) \right). \quad (155)$$

Angles use nearest periodic nodes with upward half ties, while the source's bit widths and range remain unchanged. This tie/quantization choice is an explicit specialization, not an unquoted algorithm recovered from an unavailable source ZIP. [N05]

For an angular width b , $h_\theta = 2\pi/2^b$. A nearest-node quantization gives $|\delta\rho| \leq h_\rho/2$ and circular $|\delta\theta| \leq h_\theta/2$ within the chosen range. Using the exact source radial step, a conservative planar error bound is

$$\|\hat{x} - x\| \leq r_h(e^{h_\rho/2} - 1) + 2r_h e^{h_\rho/2} \sin(h_\theta/4). \quad (156)$$

It follows from adding a radial displacement and an angular chord at the enlarged radius. The bound scales with radius; log packing is not uniform physical resolution. [U, pp. 4, 8–9]

30.1 Default diagnostic chart

The demo chooses $r_0 = 100000$ m and a 0.1 m origin core. It keys only values with $-20 \leq \rho \leq 0$. These choices define an indexed diagnostic domain, not the receiver's positioning accuracy or a geographic route. Out-of-range or core values keep the full position and a chart status without wrapping them into a new position.

30.2 No equal-information compression claim

U explicitly audits the ratios 192/64, 32/1 and 512/12 as nominal width ratios between records with different semantics. This specialization does not reintroduce a claim that one address is equivalent to all original floating information. [U, pp. 8–11]

The packed key is useful for lookup and local indexing after the relevant state exists. Position estimation, corrected observations, uncertainty and event verification remain real computational and storage costs.

31 Linear time, winding and separate topology profiles

For reference tick X_0 , period $P > 0$ and full linear tick X ,

$$u = \frac{X - X_0}{P}, \quad w = \lfloor u \rfloor, \quad \psi = u - w \in [0, 1). \quad (157)$$

The source's example $X = 1920$, $X_0 = 1135$, $P = 256$ yields $w = 3$ and $\psi = 0.06640625$. The new helper reproduces that value, including floor-division behavior for negative ticks. [U, p. 5]

The satnav key stores only the 14-bit tick residue, while the event retains full tick, GPST coordinate, origin/period and winding. A period of 16384 one-second ticks repeats its residue after 4 h 33 min 4 s. It does not identify events across those repetitions without the additional time/lineage fields.

31.1 The source half-turn is not the reflective quotient

U's literal odd radial-wrap bundle applies

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \theta + \pi, -\phi_h, -o). \quad (158)$$

Its separately named reflective Klein profile applies

$$(\rho, \theta, \phi_h, o) \mapsto (\rho', \pi - \theta, -\phi_h, -o). \quad (159)$$

The second reverses angular orientation; the first is a rotation on the base circle with a changed internal state. The archive retains independent helpers and tests, matching the distinction in U's Figure 4. [U, pp. 5–6]

Neither rule is applied to the GNSS receiver's ECEF position, a satellite orbit or the physical Earth. A local index experiment may select either profile explicitly; it does not make a geographical discontinuity an actual displacement. [N08]

The source's finite grammar and radix refinement remain local state/index mechanisms. Their existence does not turn one receiver solution into an unbounded tree or prove that every possible trajectory is retrieved in constant time.

32 GPU kernel: one independent solve per thread

The kernel takes a bounded batch of immutable epoch metadata and a structure-of-arrays observation buffer. One thread evaluates one full independent four-state solution. Each thread owns its QR workspace, iterative estimate, status and residual record. There is no cross-epoch update or unordered reduction of position states.

```
const size_t e = size_t(blockIdx.x) * blockDim.x + threadIdx.x;
if (e < count) {
    output[e] = solve_epoch(epochs[e], observations, e, config);
}
```

The launch uses 128 threads per block. Repeated jobs can supply thousands of independent epochs; a single live low-rate receiver does not by itself provide enough parallel work to establish a GPU speed advantage. This kernel is a batch/replay specialization.

32.1 Numerical implementation

The actual shared implementation is in `include/satnav_core.hpp`. It performs the retained ASA arithmetic, row construction, QR update, convergence and final covariance/residual calculation. The CUDA translation unit invokes that implementation with its device reader. The CPU uses the same word and numerical function with a plain double reader.

An independent Python implementation uses separate matrix construction and Householder QR. It is not merely calling back into the shared C++ function. The two verification levels therefore distinguish device/host conformance from an additional numerical reference comparison.

32.2 Potential performance costs remain visible

Independent threads may finish in different iterations, producing normal warp imbalance. QR uses FP64 arithmetic and square roots; no tensor-core speed claim is made. The source keeps full precision where satellite-range subtraction requires it, rather than forcing coordinates into float textures or one-bit words.

The build disables fused contraction for the shared review profile and does not enable fast math. The acceptance checks use named tolerances, not a claim of bit-identical floating evaluation across every compiler and processor.

33 Texture and global input paths

For N independent epochs, channel $c \in [0, 31]$ and observation field $f \in [0, 5]$, flatten the input as

$$\text{index}(e, c, f) = (6c + f)N + e. \quad (160)$$

At a fixed channel/field, neighboring epoch threads therefore read neighboring values. The six fields are satellite x, y, z , raw code, additive correction and nominal sigma.

33.1 Exact FP64 bit transport in the texture path

A CUDA linear texture resource uses an `int2` channel. Each texel carries the 64 raw bits of one double; the device reconstructs the value using the low and high 32-bit words. The read uses an integer `tex1Dfetch`; there is no interpolation, normalized-coordinate lookup or conversion of ranges into float values. [H4][N10]

```
const int2 bits = tex1Dfetch<int2>(texture, integer_index);
const double value = __hiloInt2Double(bits.y, bits.x);
```

The global path reads the same double buffer directly. Both paths use the same solver and output record. Equality of decoded inputs is a data-representation property; it does not guarantee a texture speedup for this access pattern.

33.2 Ownership and publication

Device buffers and events use local ownership wrappers. A texture object is destroyed before its backing buffer. Each launched epoch has one writer. The program checks CUDA allocation, copies, launch and completion before publishing downloaded result files. A failed device call is an error, not a CPU fallback labelled as a GPU success.

The first chunk receives an untimed warmup. Device events measure subsequent kernel intervals and their sum is reported. CSV parsing, allocations, texture construction, host packing, uploads, downloads and the Python event adapter are outside that sum. A CPU loop interval and a full application latency are therefore distinct quantities.

34 The RTX 5070 Ti Laptop target and VRAM plan

NVIDIA lists the **RTX 5070 Ti Laptop GPU** with a **12 GB GDDR7** memory configuration. NVIDIA's CUDA table lists the GeForce RTX 5070 Ti family at compute capability 12.0; the build selects `sm_120` and matching virtual code. CUDA 12.8 introduced compiler support for `SM_120`. These are supporting implementation facts, not equations supplied by the UGTS prototype. [H1–H3]

The program's `-probe` path records the actual device name, major/minor capability, driver/runtime versions, total/free memory and linear-texture limit. This edition's targeted binary reports a mismatching capability rather than assuming that a marketing label guarantees the compiled architecture.

34.1 Payload accounting

The executed CPU build reports 72 bytes per epoch metadata record and 376 bytes per result. The fixed observation representation is $32 \cdot 6 \cdot 8 = 1536$ bytes per epoch, so the tested ABI payload is

$$M_{\text{payload}}(N) = N(72 + 376 + 1536) = 1984N \text{ bytes.} \quad (161)$$

The 256-epoch demo therefore uses 507,904 payload bytes. A maximum chunk of 65,536 epochs uses 124 MiB before a separate 64 MiB planning margin. These figures do not count the entire driver/context or all host copies.

Admission uses the actual free/total bytes, the user budget and reserve. The defaults are a 512 MiB budget and 1536 MiB total-device reserve. After checking that each subtraction is valid, capacity is bounded by

$$B_{\text{usable}} = \min(B_{\text{user}}, 0.7F, T - R_{\text{reserve}}) - 64 \text{ MiB.} \quad (162)$$

The integer implementation rounds the free-memory fraction downward, caps each chunk at 65,536 epochs and checks texture/index limits. Another process may consume memory after probing, so actual allocation errors still propagate.

The package does not fill all 12 GB to demonstrate activity. Useful payload depends on the batch; automatically leaving space for other laptop processes is a resource plan, not a claim of permanent texture-cache residency. No operating-system ring-0 driver or executable-code-in-texture mechanism is part of the CUDA interface.

35 Actual input and result files

The executable reads two CSV files with exact headers. The source-to-domain model is recorded separately in `profile.json`; prepared real data must supply its own correction/product provenance.

```
epochs.csv:
epoch_id,t_rx_gpst_s,x0_m,y0_m,z0_m,b0_m,
asa_mask,na_mask,boundary_mask

observations.csv:
epoch_id,channel,satellite_id,sx_rx_m,sy_rx_m,sz_rx_m,
code_m,add_correction_m,sigma_m,ready
```

The display above wraps the headers for readability; the actual files use one header line each. `docs/INPUT.md` defines their exact names, ranges and physical meaning. The reader rejects malformed headers, duplicate channels, nonfinite fields and invalid sigmas before passing data to a backend.

35.1 Results

`solutions.csv` retains one row per input epoch: source/time identity, solver and fit status, iteration count, channel count, all four ASA fields, position/clock, formal diagonal covariance and residual statistics. `residuals.csv` records each active post-fit channel when the solver converges.

For a failed solve, estimate fields may contain the starting guess or a partial iterate. The status is authoritative; those fields must not be published as a valid fix. Zero variances in an unevaluated result do not mean a perfectly known position. The parser and status record prevent that interpretation without hiding the original calculation state.

`run.json` records the backend, timing scope, selected numerical settings and actual device report where available. `ugts_events.jsonl` and `ugts_summary.json` are produced by the explicit post-processing step. They do not replace the original solution or residual files.

A new output directory is required for each run. Completed runs can be compared without silently overwriting earlier evidence. Corruption of a source file or a changed result is visible in the delivered integrity manifest or retained event chain.

36 Ordered coupled records and source lineage

A CGK-R1 event binds the original solution and residuals, modeled position and velocity, executed J/K transition, complete native trace row, both compact keys, full time and prior digest. A condensed structural sketch is:

```
{
  "version": "3.6.1.7", "profile": "CGK-R1",
  "trajectory_id": 0, "epoch_id": 0, "input_index": 0,
  "event": "COUPLED_OBSERVATION_ADVANCED",
  "observation": {"solver_record": {}, "epoch_record": {}},
  "residual_records": [], "formal_variance_m2": {},
  "modeled": {"enu_m": [], "ecef_m": [],
    "velocity_enu_m_s": [], "q": 0, "hoop_phi_rad": 0},
  "jk": {"before": 0, "j": 0, "k": 0, "after": 0},
  "native_state": {"complete_executed_trace_row": "..."},
  "chart": {"up_m": 0, "time_gpst_s": 0, "winding": 0},
  "packing": {"contiguous_hex": "...", "morton_hex": "..."},
  "lineage": {"sources": {}, "native": {}},
  "previous_hash": "...", "record_hash": "..."}
}
```

This is a structural sketch; the delivered exports contain full numeric arrays, source records, statuses and hashes. Unavailable observations produce explicit prediction events; a failed mechanical advance produces a failure event. The modeled state is never presented as the original receiver measurement.

The chain uses a domain-tagged SHA-256 of canonical local JSON and retains the prior head. It binds the emitted sequence to a retained reference. It is not a signature, a satellite-signal authentication primitive or proof that an upstream observation was physically true.

Author attribution is included as supplied: Tom Klootwijk, with the identifier and date in the release metadata. A full profile/source digest remains separate from the 64-bit coordinate key or a one-bit event field. This preserves the base engine's identity/state distinction and U's requirement that uncertainty and lineage survive packing. [A, retained M2 project contract] [U, pp.8–10]

37 The retained deterministic satnav batch

The shipped demo has 256 artificial receiver epochs. The anchor is a declared WGS84 location at 52° latitude, 5° longitude and 100 m ellipsoidal height. Receiver points are generated from a local ENU path, with a known clock bias near 75,000 m. Satellite directions and ranges are constructed fixtures, not a downloaded constellation or an observation session from the user's laptop.

The first 128 cases are noiseless model data. The next ordinary cases add independent Gaussian fixture noise with their specified sigmas. The final six cases deliberately exercise failure and edge conditions. Corrections are labelled artificial additive test values, not an atmosphere or ephemeris model.

Outcome	Observed count / result
Converged numerical epochs	253 of 256
Too few selected satellites	1
Whole-word absorbed	1
Rank-deficient geometry	1
Converged within chosen residual budget	251
Gross-outlier residual alert	1
Exactly four, no residual redundancy	1

The exact-four and 32-channel cases verify those interface boundaries. The outlier case adds an artificial 200 m error to one code observation. It converges to an incorrect candidate with a large residual alert, demonstrating why convergence and navigation reliability are not the same field.

The earlier observation-only geometry-support adapter emits 251 verified candidate records, three no-solution records and two unresolved candidates under its declared conditions. The exact-four case remains unresolved because no redundancy is present; the outlier remains unresolved because the chosen residual criteria are exceeded. CGK-R1 instead advances 253 observation-driven and three prediction-only states; residual classifications remain diagnostics beside the original solutions.

These are finite results of a known model. The demo is useful for reviewing signs, units, frame adapters and implementation agreement. It is not a real-world accuracy measurement or a GPU speed result.

38 Independent results and error measures

The C++ solve is compared against a separately written Python Householder solver. The comparison checks epoch identity, all mask fields, status/fit, position and clock, formal covariance and residual statistics, yielding **4,828 checked comparisons** over the 256 epochs.

Cross-reference quantity	Observed maximum
Absolute state-component difference	$1.210719347 \times 10^{-8} \text{ m}$
Formal covariance diagonal difference	$6.394884622 \times 10^{-14} \text{ m}^2$

The comparison tolerance for each state component is 10^{-4} m . Small algorithmic agreement does not make the underlying physical inputs exact; it shows that the two implementations solve the same declared finite problem to closely matching values.

Fixture class	Epochs	3D position RMSE	Maximum error
Noiseless	128	$3.61 \times 10^{-9} \text{ m}$	$7.84 \times 10^{-9} \text{ m}$
Noisy	122	1.6017 m	3.3883 m
Gross outlier	1	147.0987 m	147.0987 m
Exactly four	1	$1.23 \times 10^{-8} \text{ m}$	$1.23 \times 10^{-8} \text{ m}$
32 channels	1	$5.71 \times 10^{-9} \text{ m}$	$5.71 \times 10^{-9} \text{ m}$

Errors above are measured only against the synthetic generating truth. The nanometre- scale noiseless figures reflect arithmetic roundoff in matched artificial data, not an achievable GNSS hardware accuracy. The noisy fixture demonstrates the selected noise model; a real receiver's error includes its actual correction/model limitations.

Formal covariance, predicted geometry quality and empirical truth error are different quantities. The package keeps them separate, following the distinction made in the positioning-error reference. [G5]

39 Prior release verification record

This section retains the 3.6.1.5 preparation record. Archived 3.6.1.6 evidence is given in the SRK-R1 validation section; historical 3.6.1.7 checks appear in the coupled implementation validation section. Retained 3.6.1.8 evidence appears in Section 3.9. The following unexecuted CUDA statements refer to the prior preparation environment only.

Category	Recorded result
CPU C++ build and CTest	Passed
C++ explicit checks	105,559 passed
Python UGTS/geodesy tests	18 passed
Command-line scenarios	11 passed
Independent solver comparisons	4,828 passed
CPU Address/UndefinedBehavior Sanitizer	CTest passed
CUDA compilation	not_run: nvcc unavailable
CUDA execution / memory sanitizer	not_run: no accessible NVIDIA GPU
Windows native build	not_run in preparation
Live GNSS observation validation	Not performed; synthetic fixtures only

The C++ count includes all 65,536 possible 16-bit population-count cases and 10,000 independent random word-mask comparisons. It also checks the full JK truth table, QR identities, covariance, rank rejection and the retained whole-word rule. A large assertion count is not a count of separate receivers tested.

The Python suite checks both key codecs against the source's golden pair and 5,000 random tuples, all prefix lengths for the reference key, quantization bounds, full winding, two separate topologies, cone/sweep relations, the jitter-interval distinction, geodetic/ENU round trips and literal OTAN2 semantics.

The CLI scenarios include malformed IDs/channels, negative sigma, NaN input, invalid back-end/options, existing-output protection, a normal run and an iteration-limit run. These prior outcomes remain in the preserved sibling releases, with earlier source test claims kept separate from these actual executions.

In the prior preparation, the device validation script was invoked and recorded its unavailable-tool result. Later releases supply separate actual laptop execution records. No general CUDA speedup or texture-residency claim is made in this edition.

40 Build, run and verify on the requested laptop

Use a CUDA-supported host compiler, CMake 3.24 or newer, and CUDA Toolkit 12.8 or newer for the requested SM120 target. The installed NVIDIA driver must support the chosen runtime/-toolchain. This specifies a minimum architecture-capable toolkit, not a claim that 12.8 is the newest available release. [H2, H3]

```
cmake -S . -B build_gpu -DSATNAV_ENABLE_CUDA=ON \
  -DCMAKE_BUILD_TYPE=Release
cmake --build build_gpu --config Release --parallel 2
ctest --test-dir build_gpu -C Release --output-on-failure
./build_gpu/satnav --probe
./build_gpu/satnav --input examples/demo --out run_gpu \
  --backend cuda --read texture --verify
python tools/verify_run.py --input examples/demo --run run_gpu
python tools/ugts_handoff.py --input examples/demo --run run_gpu
```

On a Windows multi-configuration build, use the following executable path:

```
build_gpu/Release/satnav.exe
```

Run configuration from the matching x64 compiler developer terminal so that CMake can locate the host toolchain.

40.1 Complete device comparison

```
python tools/validate_gpu.py --sanitizer
```

The script records compilation, device probing, texture/global runs, shared CPU comparisons, independent Python comparisons and requested memcheck results. It uses fresh run directories. Missing required tooling produces a separate unavailable status and exit code 3. A real build/test error remains a failed status.

40.2 CPU-only review

```
cmake -S . -B build_cpu -DCMAKE_BUILD_TYPE=Release
cmake --build build_cpu --config Release --parallel 2
./build_cpu/satnav --input examples/demo --out run_cpu
python tools/verify_run.py --input examples/demo --run run_cpu
python -m unittest discover -s tests -p "test_*.py" -v
```

The retained SATNAV reference uses the Python standard library. The coupled reference additionally requires NumPy, installed from `requirements.txt`.

41 Integration decisions and source issue register

ID	Decision	Effect
N01	Corrected-code satnav model	The GNSS observation equation and solver are explicit new additions, not attributed to U.
N02	Full receiver state retained	Solve in 3D ECEF plus clock; use a separate local log chart and retain Up and provenance.
N03	Independent QR methods	Shared CPU/CUDA Givens implementation and independent Python Householder solve; named rank and convergence policy.
N04	Fixed channel-to-bit mapping	One epoch is one at-most-32-channel word. Boundary absorption stays whole-word.
N05	Source key layouts preserved	Exact contiguous/Morton pair retained; quantization ties named; key does not replace full state.
N06	Two jitter intervals separated	Explicit source equality issue documented; perturbation-family and measurement-centred intervals are not silently equated.
N07	Bounded sweep interpretation	Translation-only analytic interval; floating arithmetic and nonnested grids not promoted to stronger certificates.
N08	Topology does not rewrite Earth	Bundle/reflection helpers remain separate local contracts, not GNSS propagation rules.
N09	Computational event semantics	Declared support/residual policies produce observation records, not a protection level, spoof-proof fix or actuator command.
N10	Recorded hardware evidence	CUDA was unexecuted in 3.6.1.5 preparation; current device builds, replay and memcheck results are recorded separately.
N11	Explicit literal feedback	SRK-R1 composes ASA/NA and JK with supplied Boolean expressions and synchronous state carry.
N12	Complete geometric coupling	CGK-R1 defines circle-plus, predicate encoders, state-dependent mechanics and eigenmatrix; the actual handoff executes and publishes the coupled state.
N13	Live receiver positioning	LIVE-GPS-L1-R1 decodes real GPS observations and streamed ephemerides, applies explicit physical corrections and solves the remote receiver position during capture.

These decisions define the explicit profiles of aTOMos 3.6.1 with the retained UGTS prototype integrated at explicit interfaces. They do not discard the user's operator vocabulary or pretend that all missing GNSS physics were already encoded in a name.

A later variant can extend the estimator with new states, observable types or correction adapters. It must identify the changed observation model and preserve a projection or compatibility test to this profile where appropriate. An unreported inter-system bias, new residual exclusion rule or changed topology is not a transparent optimization.

42 Source and artifact inventory

U. *UGTS_KC_3_6_2_SCLP_Tom_Klootwijk(1).pdf*. The file has 19 physical pages and a 16-page numbered main/appendix sequence. The source's referenced 13-page precursor and software ZIP were not supplied. The inherited source manifest records its PDF SHA-256 as

```
6987490a0a7b7ed76b0091057a2e69ab831fee786434a0b4219782923f5ea580
```

The U page references in this formalization point to the printed main numbers. For example, its key table on printed p. 8 is physical PDF page 11. The relevant figure, table and jitter-equation pages were inspected as rendered pages as well as text.

A. aTOMos v3.6.1 / M2 source and code contract in this project conversation. The prior archive was not mounted in this turn; the specialization preserves the written ASA/JK/OTAN2 contracts rather than claiming a binary patch to an unavailable artifact. The actual new source is included and tested to the extent recorded above.

Delivered file family	Purpose
include/, src/, cuda/ python/	Shared solver, executable and real CUDA backend source. Independent solution and source-derived UGTS/geodesy helpers.
examples/demo/ tests/, results/ tools/	Prepared artificial epochs, observations, truth and profile. Reproducible checks and the actual recorded outcomes. Build/review, independent comparison, event adapter and integrity tools.
docs/	This editable formalization, input contract and source-decision register.
source/manifest.json	Exact source availability and fingerprint record.
SHA256SUMS.txt	Delivered-byte integrity manifest.

Source font files and proprietary binaries are not distributed. The current uploaded UGTS reference may be retained separately for source comparison; its equations and selected mechanisms are integrated into this volume instead of concatenating unrelated older transcripts into the kernel package.

Primary implementation references

Recorded by the 3.6.1.5 preparation, 14 September 2026. These external references support the satnav model and hardware contract; they are separate from the user's source corpus.

- G1 ESA Navipedia, J. Sanz Subirana, J.M. Juan Zornoza and M. Hernández-Pajares, [Code Based Positioning \(SPS\)](#). Four-state model, correction terms, linearized geometry and iterative position update.
- G2 ESA Navipedia, [Satellite Coordinates Computation](#). Emission-time satellite state and reception-time frame treatment.
- G3 ESA Navipedia, [Transformations between ECEF and ENU coordinates](#). Local frame conversion and angular conventions.
- G4 ESA Navipedia, [Parameters adjustment](#). Linearized residual and least-squares framework.
- G5 ESA Navipedia, [Positioning Error](#). Formal covariance, predicted geometry and measured error distinguished.
- G6 NGA Office of Geomatics, [World Geodetic System 1984](#). Ellipsoid/reference-system definition and constants used in the coordinate utility.
- H1 NVIDIA, [GeForce RTX 50 Series Laptops](#). RTX 5070 Ti Laptop memory specification.
- H2 NVIDIA, [CUDA GPU Compute Capability](#). GeForce RTX 5070 Ti family listed at compute capability 12.0.
- H3 NVIDIA, [CUDA 12.8 Features Archive](#). Introduction of SM_120 compiler support.
- H4 NVIDIA, [CUDA Runtime API: Texture Object Management](#). Linear resource and element-read contract.

aTOMos v3.6.1.10 — Orbital seeds and literal state

Tom Klootwijk NL200678942 10-07-1990

Profiles: ORBIT-SEED-R1, LIVE-GPS-L1-R1, CGK-R1, SRK-R1 and SATNAV-R1.

Release date: 15 September 2026.

Compact explicit state, timestamp reconstruction, two ASA/NA sets, synchronous JK, ground-station horizons and measured forecast limits.